

ORANGES POSSUMS AND ANTS

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Representation

At its most basic core representation is the singular form of communication. Where it encompasses agents use of their senses to interpret information. This idea of representation morphs into the ‘ Morphology of Representation’ within its context in design.

As addressed in Podcast 1, representation is insperable from ethics and political issues. ‘ There is always ideology embedded in your line drawing,’ - Professor Helene Frichot. I think that this connection between ethics and design is critical for the future of design. Where we aim to combat issues we have generated. The most prominent example being that of climate change. We need to be conscience of the life cycle of our designs and representation is a method of exploring this.

Something that stood out to me within the conversation about methods of representation is the analogue and digital modes of representation. While the modern age leads towards digital modelling for its representative benefits of being efficient, low resource consuming and long term economical. Analogue modelling is the closest texturally, spatially and allows interaction with the scale representation of an idea immersing within the space. Representation trancends verbal communication through the power behind its intensions.



- **Agents**

All living beings are agents. Where they complete processes, aiming for goals

- **Traces**

Patterns formed as agents preform their intended processes, these are observable.

- **Information**

Data interpreted by agents through their observation of the world. By employing their senses to process information, varying agents can access different information.

- **Meanings**

Agents generate meaning through the interpretation of information about traces. Meanings can be shared creating a dedicated process of communication.

- **Transpositions**

The interpretation of a represntation and its reconstruction. Creatives to audience, developers to engineers.

- **Marks**

Extend the possibilities for sharing meanings through utelising materials and patterns.

Representation

Gwyllim Jahn - Architectural Designer

Key representation techniques:

The major techniques empoyed by Gwyllim Jahn and his team in the design and development process for their submission to the Tallinn Arcitecture Biennale include; Physical protoypes, Hololens virtual template, Parametric models and Sketches.

Jahn addresses an absence of typical techniques such as shop drawings, due to the project being void of involved share holders. The lack of contractors, and major stakeholders lent the design process to more internal relaisation. Promoting the use of spoken representation which is ‘typically avoided as they are too high risk’.

Future of representation:

Jahn highlights both AI and the rising climate crisis as key factors in the future of design. Addressing that the longer we wait to include concerns for climate change in design, the less proactive we will able to be and the more reactive we will have to become.

While these are both undeniably intrisically linked with the future of design. The Hololens virtual template stood out as the future of representation of fabrication and constuction. Jahn’s story of the Russian concretor really spoke to this form of representation as superseding spoken word, making it accessible, compatible and more efficient form of representation.



Advice for Future Designers:

- Willing to learn new things and be adaptible to the everchanging conventions of design. Especially in terms of software abilities. Focus on the strength to quickly embrace new tools instead of specialising in one that will be superseded in the future.
- Do not focus on a “correct” path within your design education, priorities learning the tools instead of optemising yourself for overly specific employability traits.





Dale Ferguson - Theatre Designer

Key representation techniques:

Within Dale Ferguson's design development he utelises a variety of representation techniques, tailored to who is being communicated with and what needs communicating. With the conceptulisation stage of Ferguson's design process, detailed writted representation in the form of annotations on scripts initialises the conceptulisation process. However the central figure in Ferguson's techniques of representation are his physical models.

Following Ferguson's collation of inspiration through reasearch, diorama models of the set piece are created. His work on Death of a Salesman presented a model of a sporting stand completed with modeled agents for reference. Ferguson praises this technique as it accounts for 'so many disparate groups have to understand the space,". Representing a physical model allows, actors, coreographers, sound designers and fabricators to understand the form for their individual needs.

Future of representation:

Ferguson invisions a future of representation guided by digital modeling and designing software. While this improves our efficiency and can generate representations of ideas quickly he does mention the vitality of physical models for representing to varieties of people interacting with a design.



Advice for Future Designers:

- Be well read and utelise literature as a fuel for discovering ones own voice. 'Important that each generation reinterprets classics,'
- 'Your skills liberate your imagination,' Ferguson pushes for future designers to upskill themselves in 3D modeling and other representative digital software in order to become more efficient at realising their ideas. The speed at which ideas can be represented using these tools allows for futher curiosity and limits burn out.



Representation Comparison (Theatre Designer vs Architectural Designer)

Similarities:

Both Ferguson and Jahn utelise digital modelling to represent their ideas and communicate it to those contributing or interacting with their designs. Both design processes include prototyping in physical forms and the designers readily admit to the importance of interacting with a physical model.

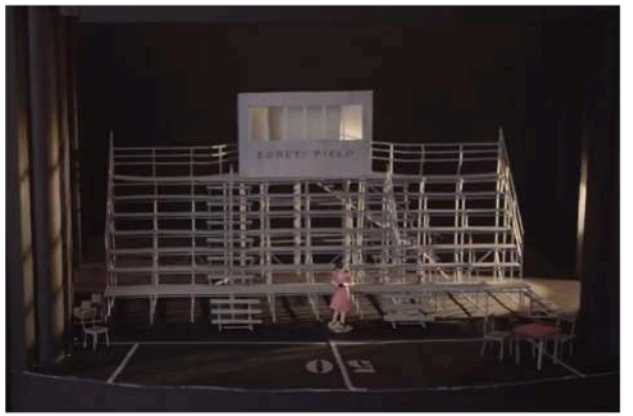
Ferguson and Jahn do not often utelise pen and paper sketches to communicate their ideas, requiring a greater depth of design to be communicated.

Differences:

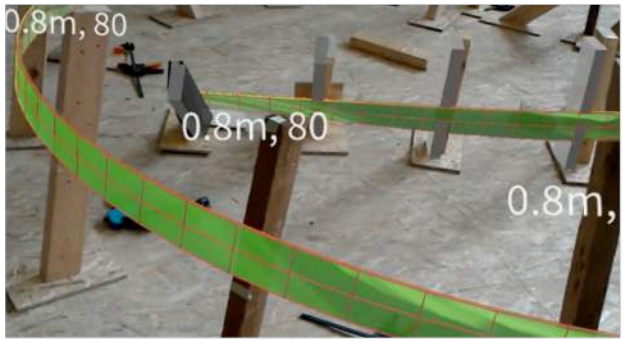
A subtle difference in the designers' methods of representating is within the order of their process. While Jahn strayed from convention by prototyping early on in his design process in order to visualise and identify how to begin modelling such a complex form. Ferguson opts to prototype later in his design process following collaboration with other contributors to the project.

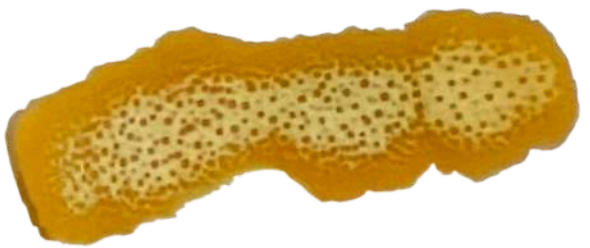
Another difference is the representation of how to construct each design. While Jahn utelises a new and complex technique of the virtual templates, Ferguson stays with the conventional tech drawings as they adequitly communicate what is required.











Representation Ethics Product Design vs Fashion Design

Both of these disciplines mold representation to target audiences and agents for their designs. These disciplines require communication to a wide variety of stakeholders, fabricators, consumers and a complex arrary of members of a design team. Within this communication, representation is the forefront for expression; aesthetically, spatially, texturally and ideologically.

All design is interconnected with the ideology of the designer and to adapt to a future world these ideologies must work in conjuction with ethical questions. Two major political issues that these disciplines must combat are:

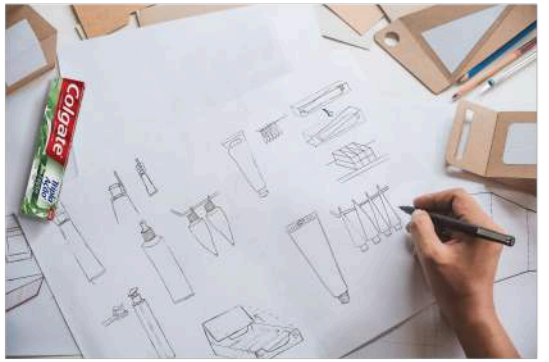
- Sustainability; with both disciplines waste managment and the shift into lean manufacturing, an understanding of a product's life cycle is needed. While both disciplines converge in their solution to design high quality and long lasting pieces fit for longevity and repeated use. Fashion design approaches this more stylistically and product design combats this constructively. Fashion design focuses on agent's interaction with trends, fighting this with the representation of timeless classics. Product design battles this through enhancing durability with both design and manufacturing techniques.

- Inclusivity and accessibility; both disciplines must acknowledge a diverse range of agents that interact with the realisation of their designs. This arrises with designing for different body types, disabilities, races and using representation in design to represent diverse groups of people.





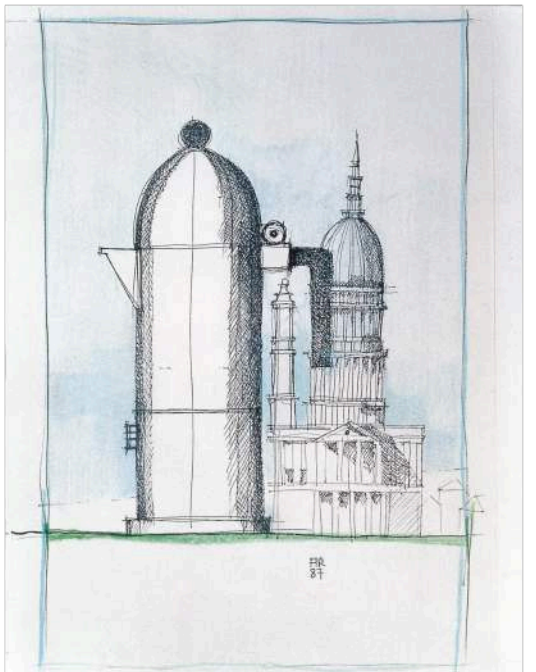




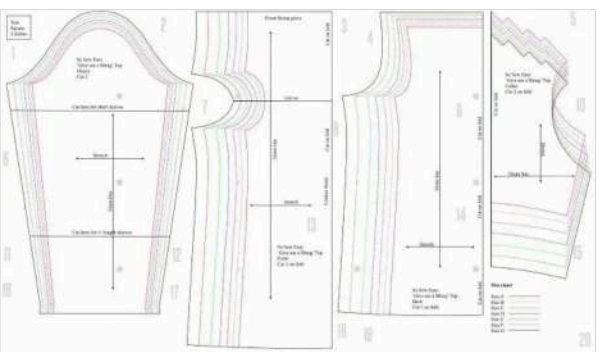
Methods of Representation Comparison

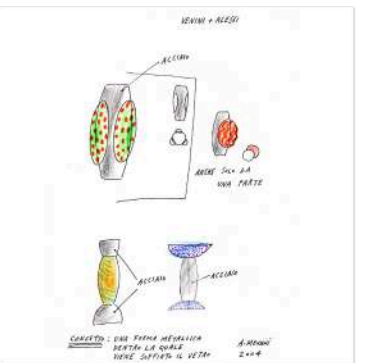
Within both disciplines inital ‘dirty’ sketching is the norm for representing idea generation and progression. However in fashion design these sketches tend to have a greater focus on artistic expression with less rigid conventions. Within product design there is a more strict perception that convention should always be conformed to, with diverging away from this convention not being as highly celebrated. However i’ve provided a counter example of this with Alberto Alessi’s early design representations. They take on a more artistic representation focusing more on colour, form and texture then dimension and user experience.

Although I previously alluded to a stark difference in the design philosophy of each discipline, somewhere they converge in order to accurately represent their designs is within technical drawings. For the purpose of manufacturing and collaboration the designs need to be represented with more explicit technical detail. In fashion design this manifests as sewing patterns and in product design as tech drawings.









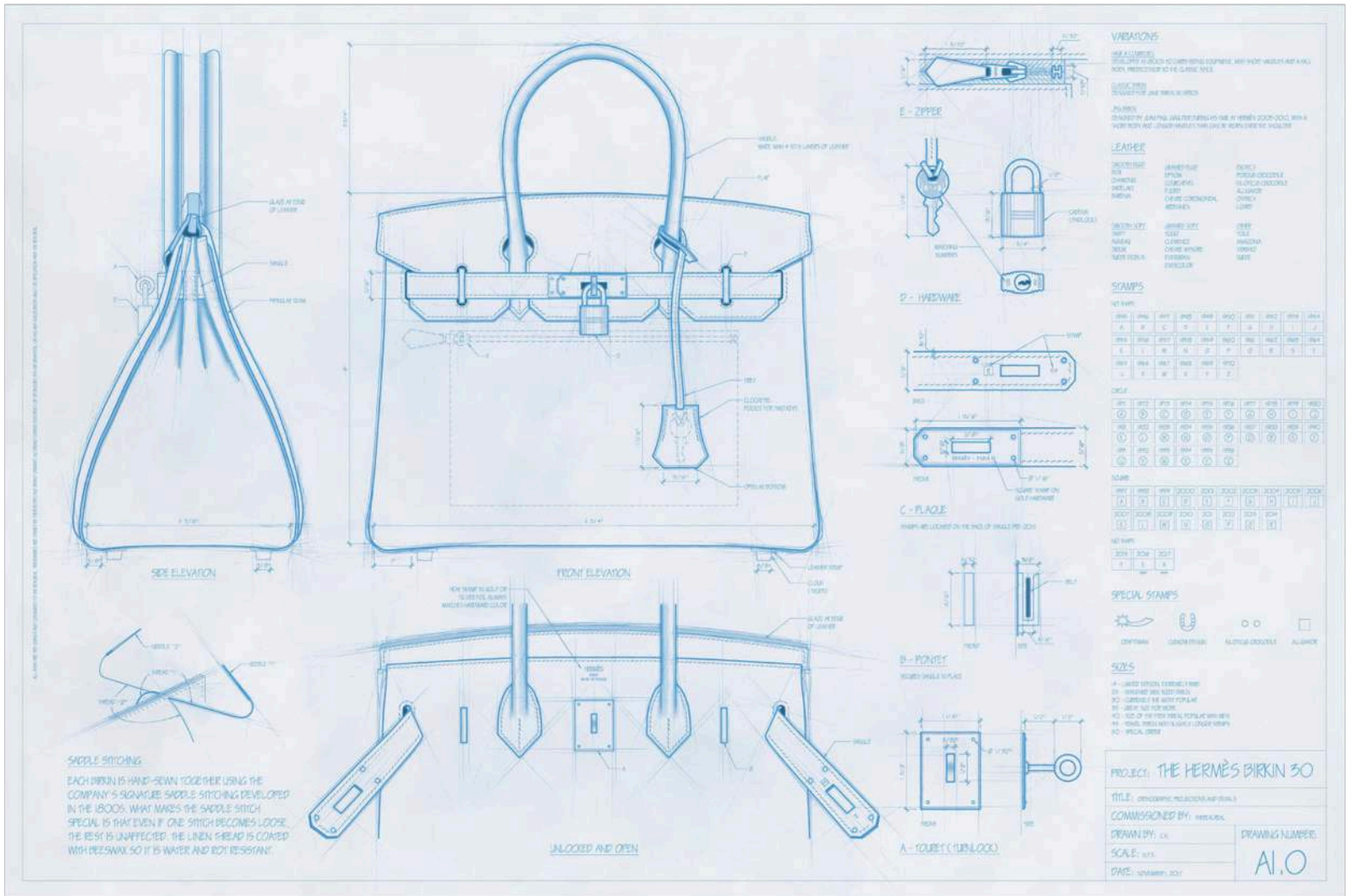


Methods of Representation Comparison

Where the two disciplines meet in my opinion is with fashion hardware and accessories. Where structure meets beauty. In the example i've previded of the Hermes Birkin bag, it is explicitly depicting complex deign details, measurement, dimension and functionality. Hardware types and movement is expressed. This is where the representation technique of technical drawings is essential.

The annotations represent the movement and purpose of the bag and aids in the designer's vision being realised on paper.





Botanical Agent

Fruit of the species Citrus x Sinensis, from the family Rutaceae.

The orange comes from the citrus x sinensis plant which is a hybrid plant of citrus maxima (pomelo) and citrus reticulata (mandarin). The orange is useful for more then its vitamin filled sweet taste, the flowers from the orange tree hold significant cultural meaning but also interact with bees to make orange blossom honey. The leaves from the tree not only provide shade for creatures and people alike but also can be boiled to make orange tea. The wood from the orange tree can be used to season grilled meat and also in the tech world as spudgers in order to manipulate electronic wires.

The most common pest to orange growth in Australia are Citrus Leafminers, but this is not the only group of agents interacting with the fruit. Bees, ants, bacteria, possums, people and robots for harvesting the fruit.

Scientific Classification

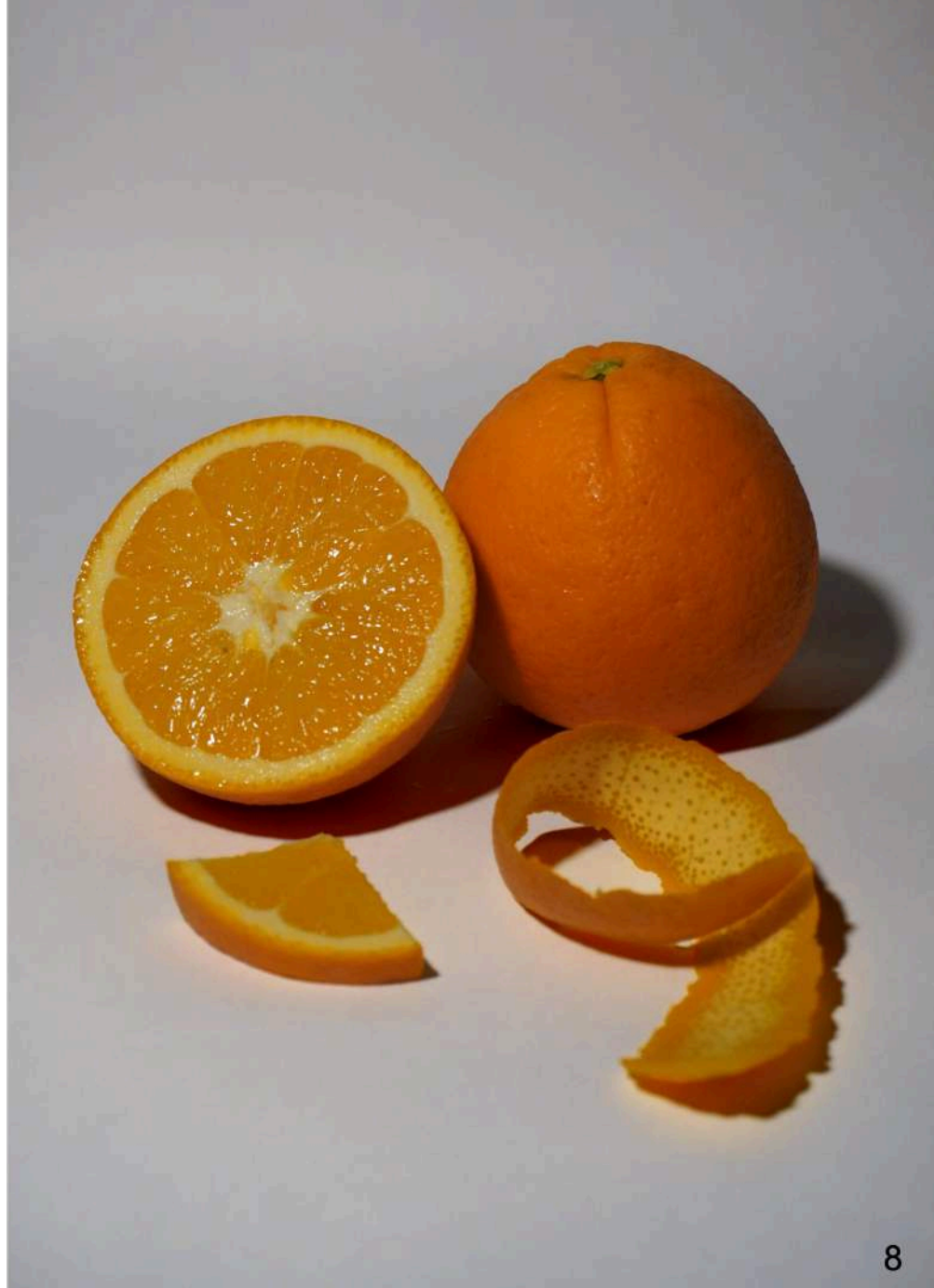
Kingdom: Plantae

Order: Sapindales

Family: Rutaceae

Genus: Citrus

Species: Citrus x Sinensis



Botanical Agent (Culture)

The significance of oranges in modern culture is tied to Chinese New Year and its symbolism for prosperity and good fortune. This symblism however is not a new concept, during the Renaissance the rarity of oranges had them become a sign of opulence.

This tie to opulence and oranges being the fruit for kings, emperors and gods forms for a long symbolic history in Greek Mythology. Hercules stealing the ‘Goldern Apple’ from the fruit’s sacred owner Venus (Aphrodite), and bringing it to the shores of ‘ Italy’ as a labor for immortality has morphed into a perception of the significance of oranges to Italian culture. Exchanging the fruit for immortality has been expressed as oranges becoming important objects of exchange.





Botanical Agent (Habitat and Community)

Habitat:

The orange originates from the area of Southern China, Northeast India and Myanmar, but I found my botanical agent in my fridge.

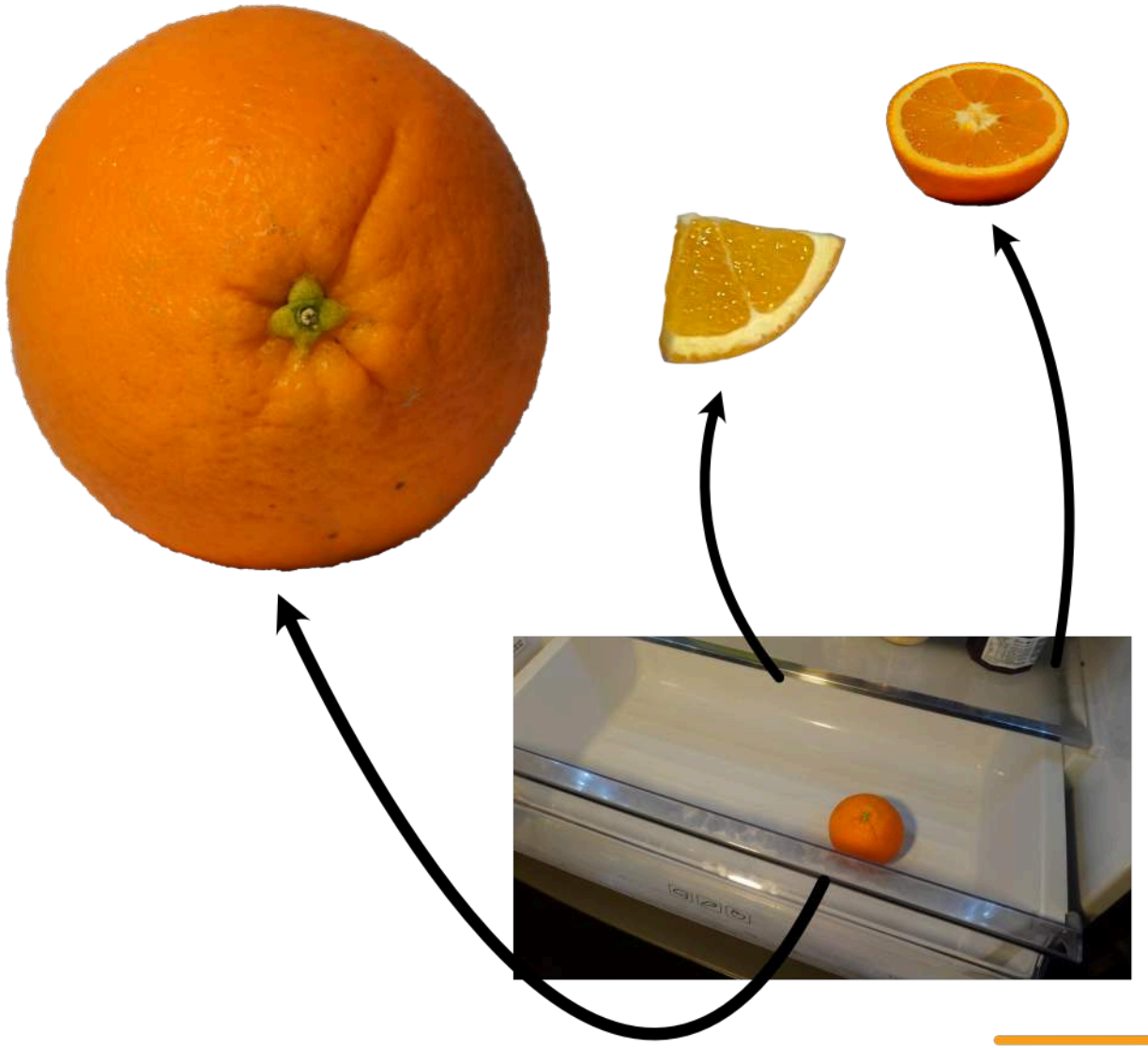
The optimal location to grow oranges is within a temperature range of 21 to 32 degrees celcius. The orange tree is considered a subtropical plant, hence why Australia's climate is optimal.

Community:

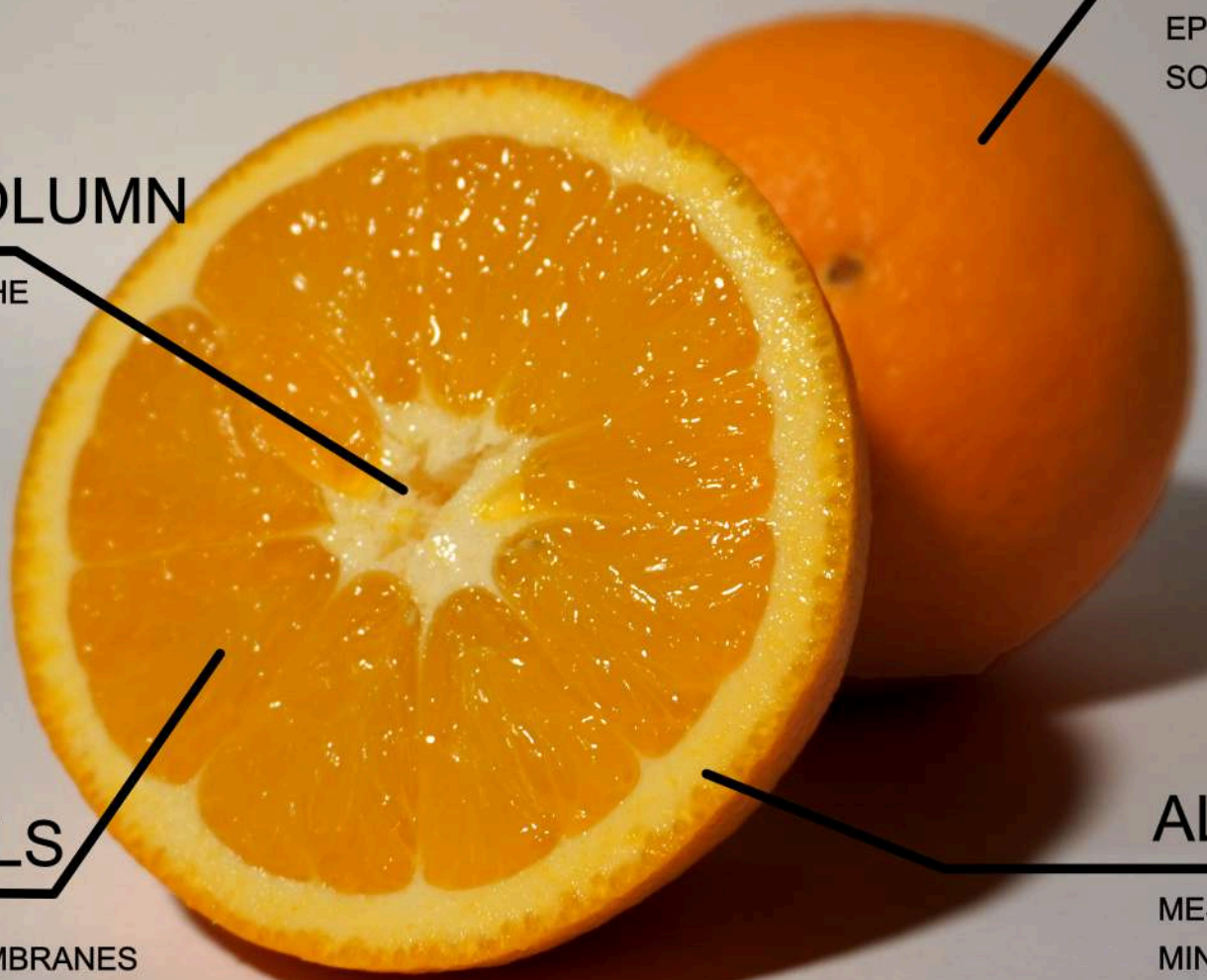
Pollinators: The flowers on orange trees can pollinate without the assistance of insects however they thrive with their help. Bees and wasps are the main pollinator of orange plants.

Predators: The predators orange trees face are typical natural predators including birds and lizards. However the sweet fruit also attacts its own uncommon predators in ladybirds, lacewing larve mites, caterpillars, rats and possums.

Parasites: The parasitic diseases facing orange tress in Australia are lemon scab, sooty blotch, melanose and Armillaria.



Herperidum [Modified Berry, Tough External Rind]



FLAVEDO

EPICARP EXTERNAL LAYER, SOURCE OF VITAMIN C

CENTRAL COLUMN

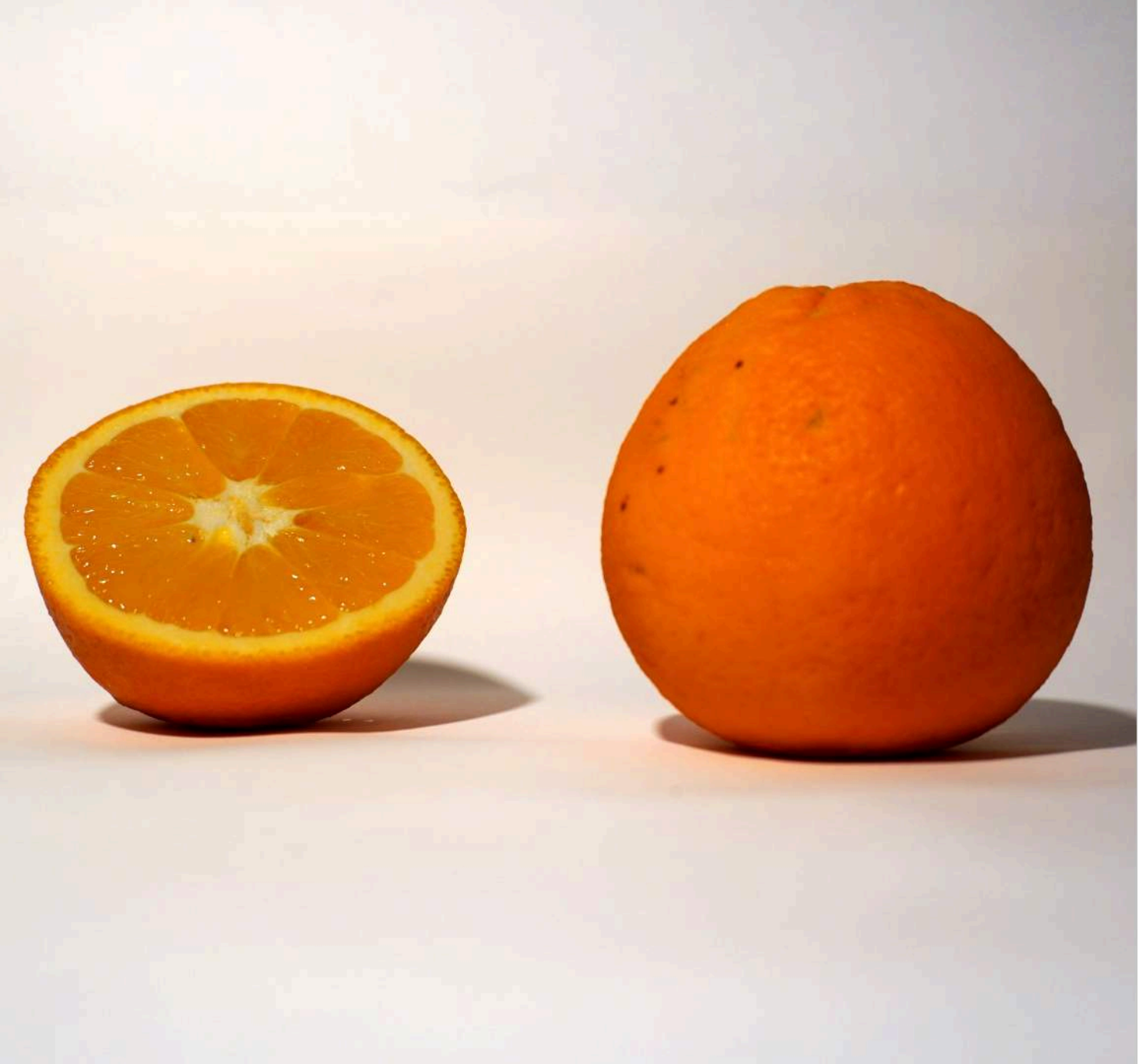
MESOCARP LAYER OF THE ORANGE, STRUCTURAL PURPOSE

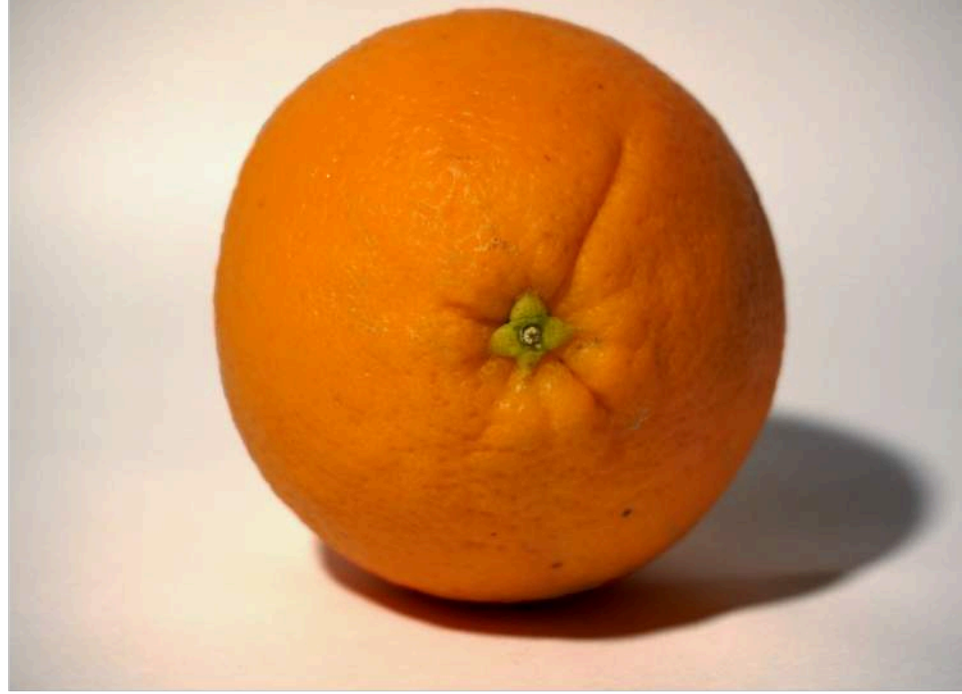
CARPELS

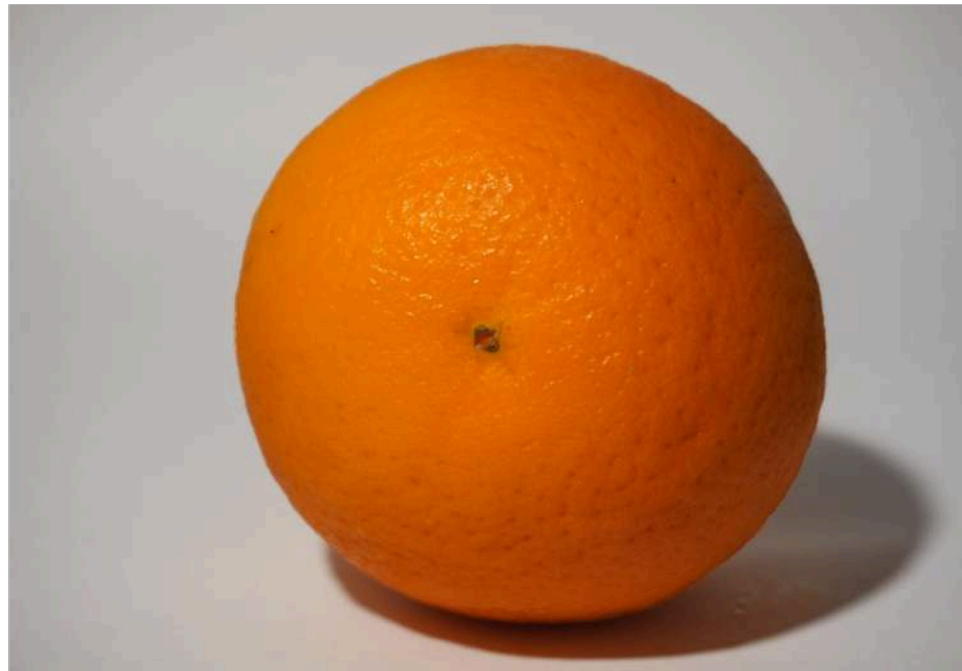
CELL LIKE MEMBRANES STORING ORANGE JUICE

ALBEDO

MESOCARP LAYER, THICKNESS DETERMINED BY THE NITROGEN AND POTASSIUM LEVELS OF THE FRUIT







Botanical Agent (History)



8 Million Years Ago:

All citrus fruits originate from the Southeast Himalayan foothills. Fossils from the area point to a common citrus ancestor.



314 BC:

The first written evidence of sweet oranges appears from Chinese lierature.



8th Century:

Sour oranges were spread by Moors following an Islamic Conquest. Where the Arab empire strenched its boarders into Central Asia, North Africa, the Middle East and Sicily. At this stange sour orange was only used in medical practises by herbalists.



940s:

The text The Meadows of Gold by al-Masudi mentions oranges having been brought to Oman. Thoughout the 10th and into the 11th century oranges continued to spread though modern day Iraq, Syria, Palestine, Egypt, Spain, Sardinia and Persia.

1475:

The first written observation of the difference between the sweet and sour orange is recorded in 1475. This is documented in a manuscript by Bartolomeo Platina for Pope Sixtus IV. This manuscript is archived in Savona Italy.



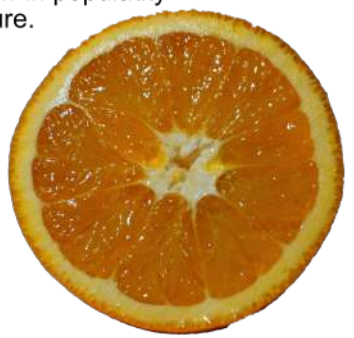
1493 - 1579:

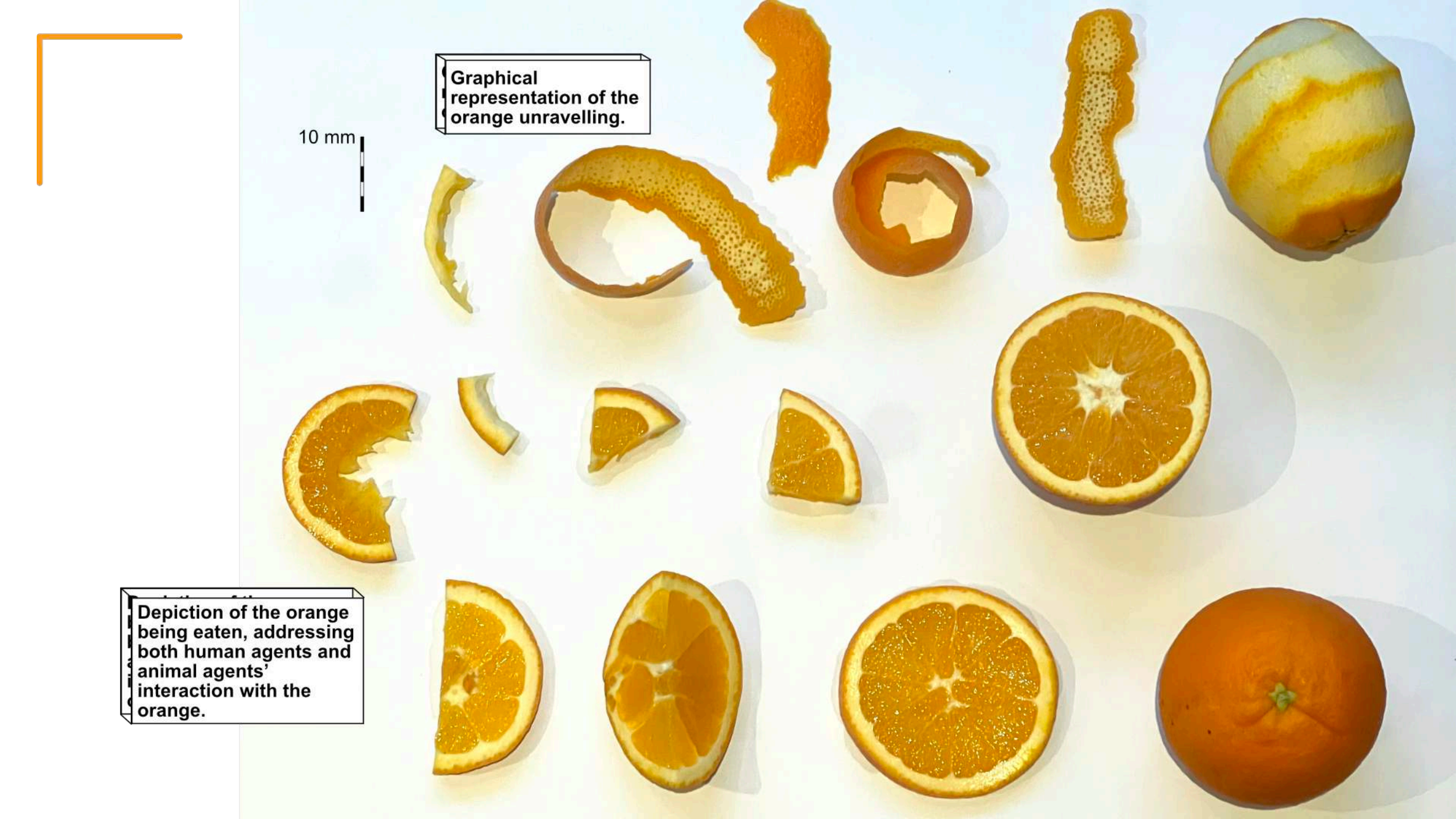
Christopher Columbus brought sweet orange seeds on his voyage to the 'New World' labeled as overseas survival tools. The seeds were planted in Haiti, Florida, and St. Augustine which was the first European settlement in America.

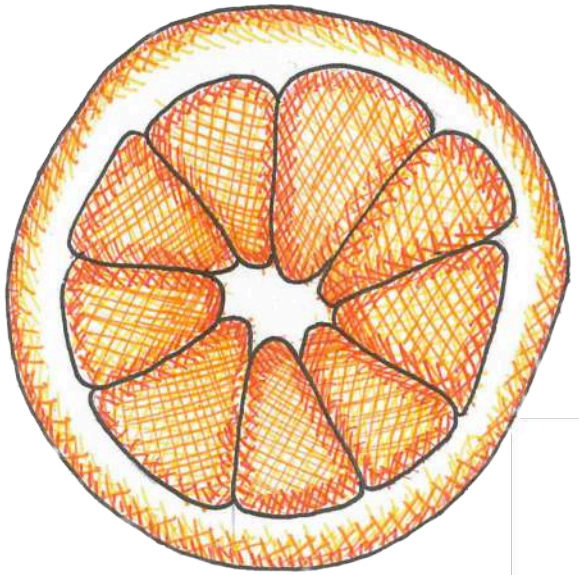


117th Century - Present:

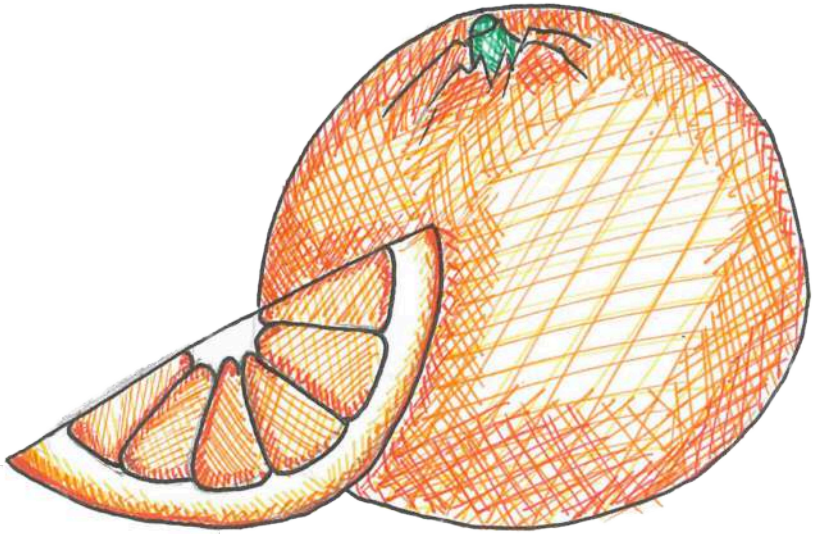
Through further global exploration oranges spread further into the Mediterranean, America, Europe and following the English settlement here in Australia was brought across the sea. Oranges grew in populatity though their famed scurvy cure.




Graphical representation of the orange unravelling.
10 mm
Depiction of the orange being eaten, addressing both human agents and animal agents' interaction with the orange.

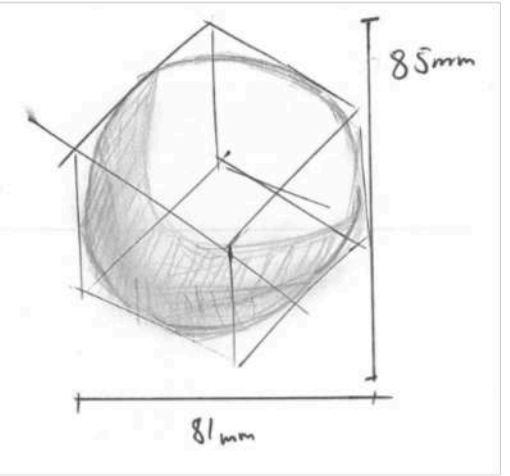


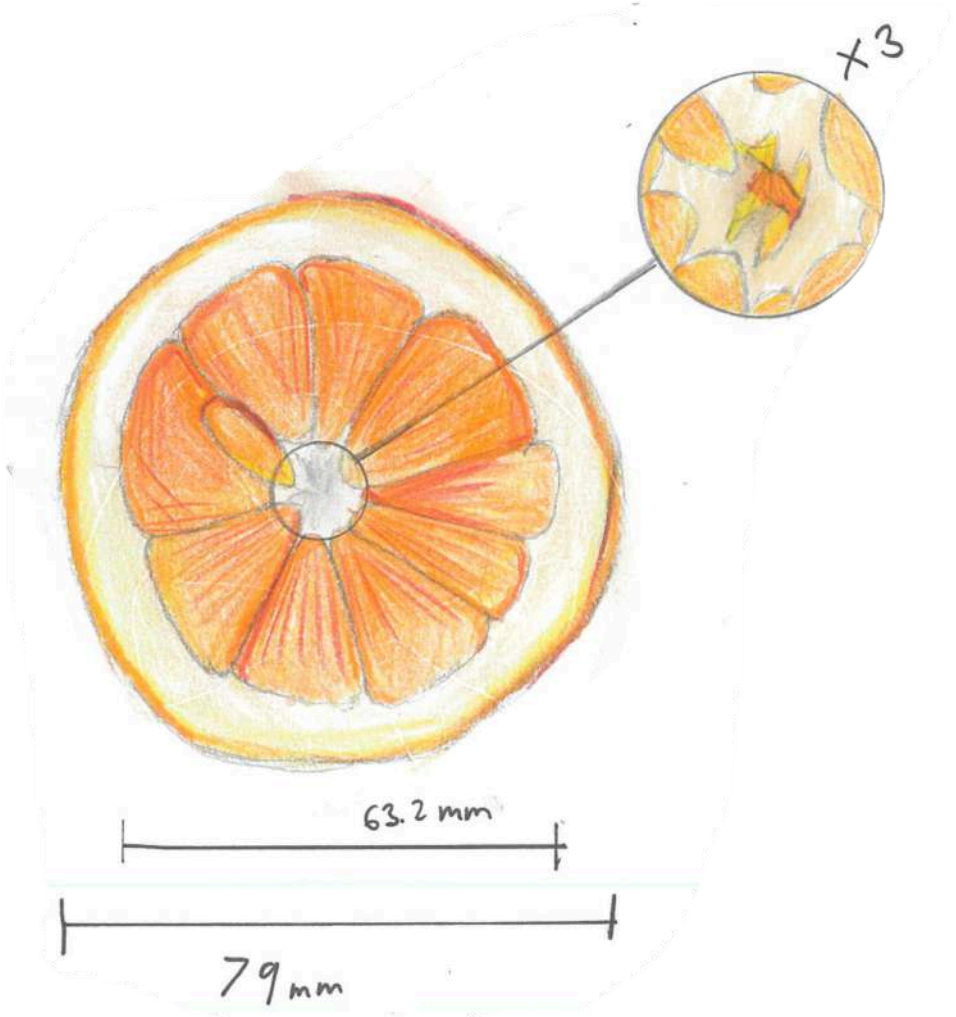
Hatching technigue with coloured pens to depict the gradient of colours and represent a more structured drawing.

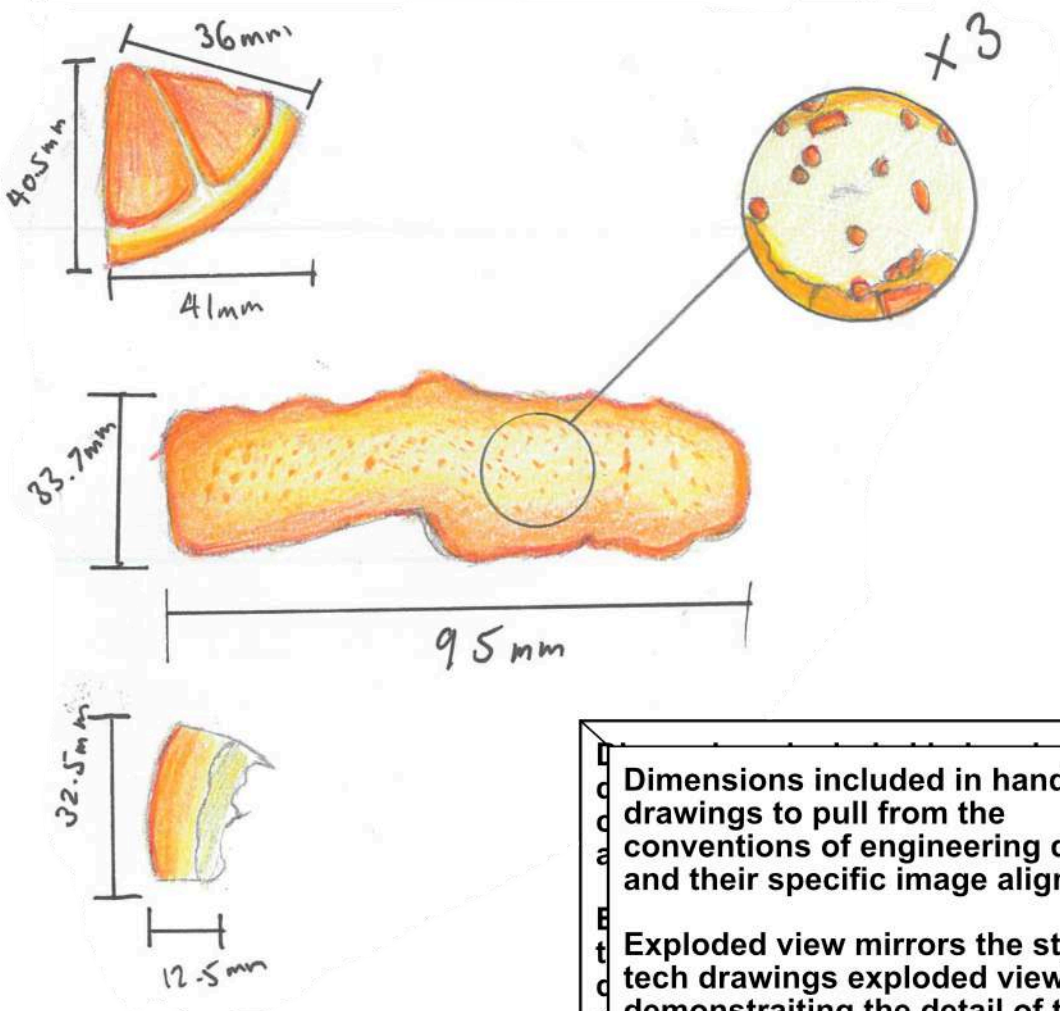




Exploration of different techniques of representing the orange. Opposing the conventional style in my discipline of engineering.

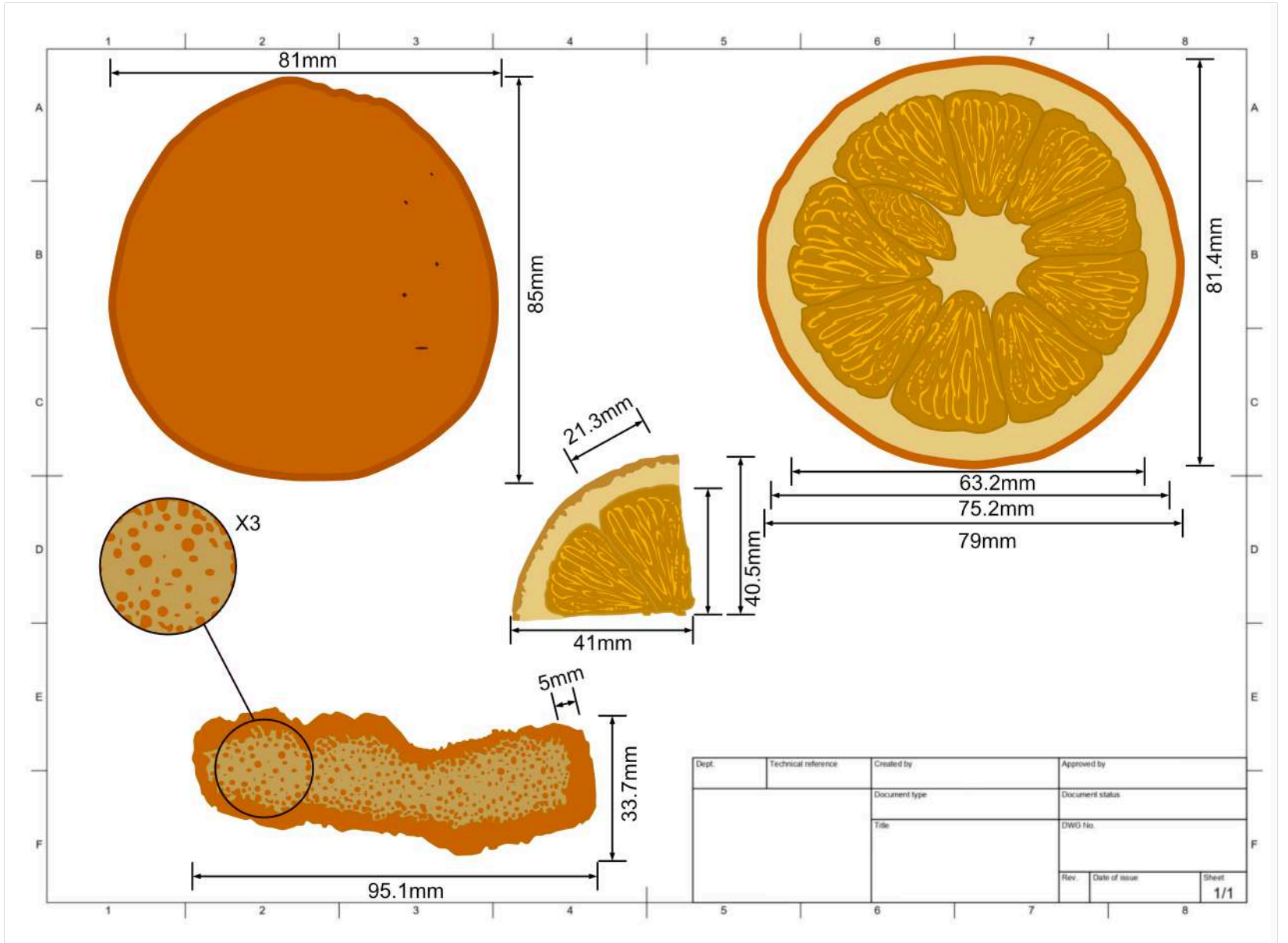




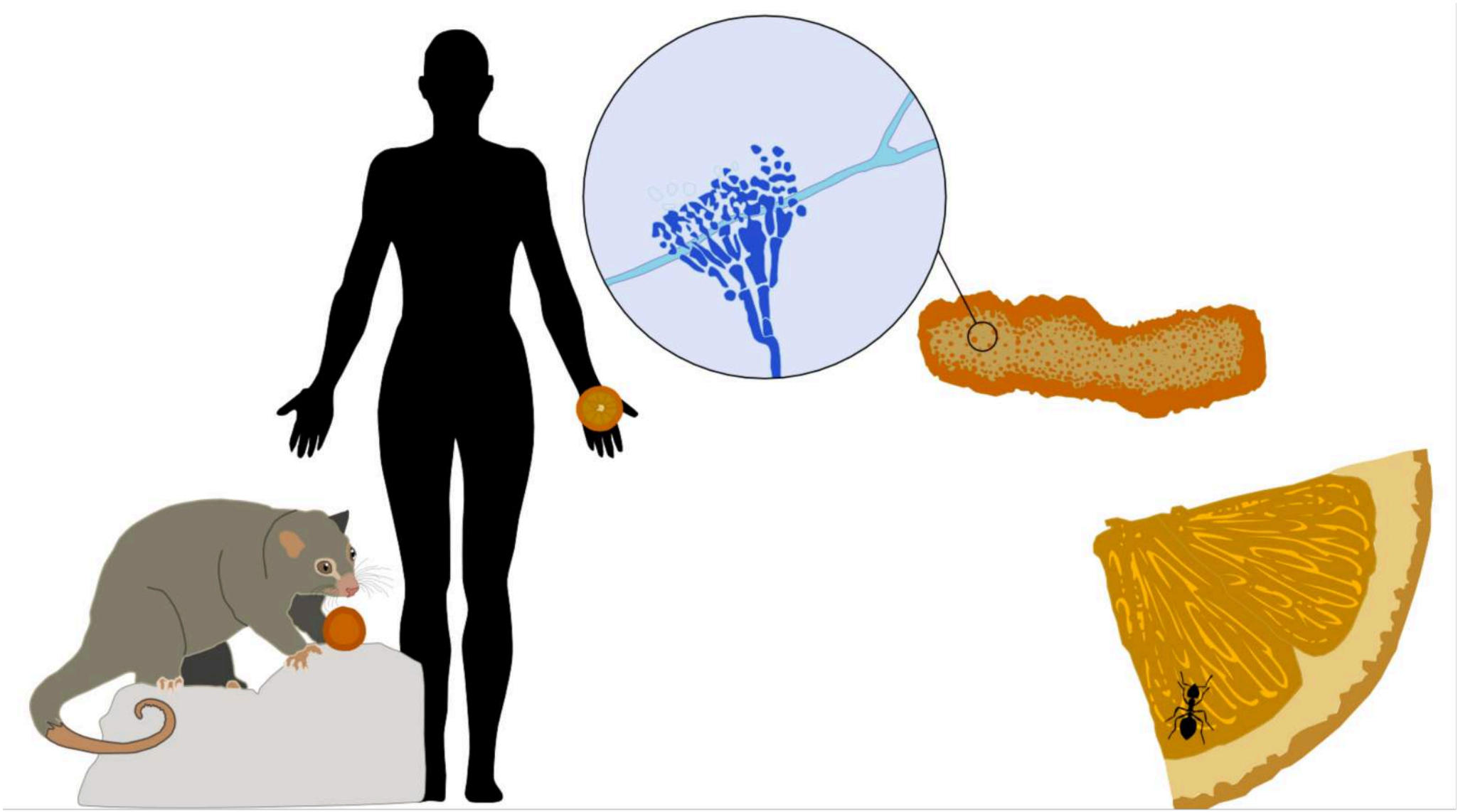


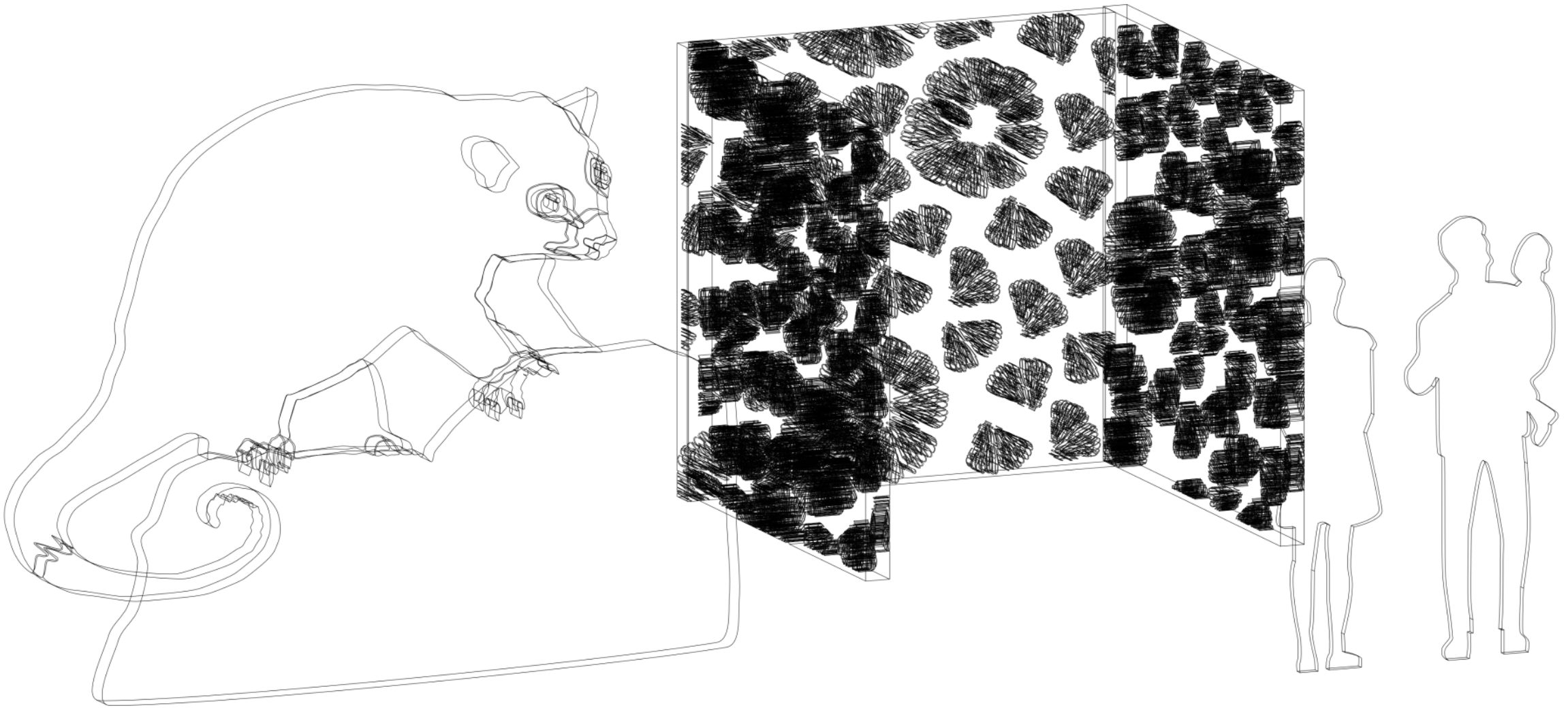
Dimensions included in hand drawings to pull from the conventions of engineering drawings and their specific image alignment.

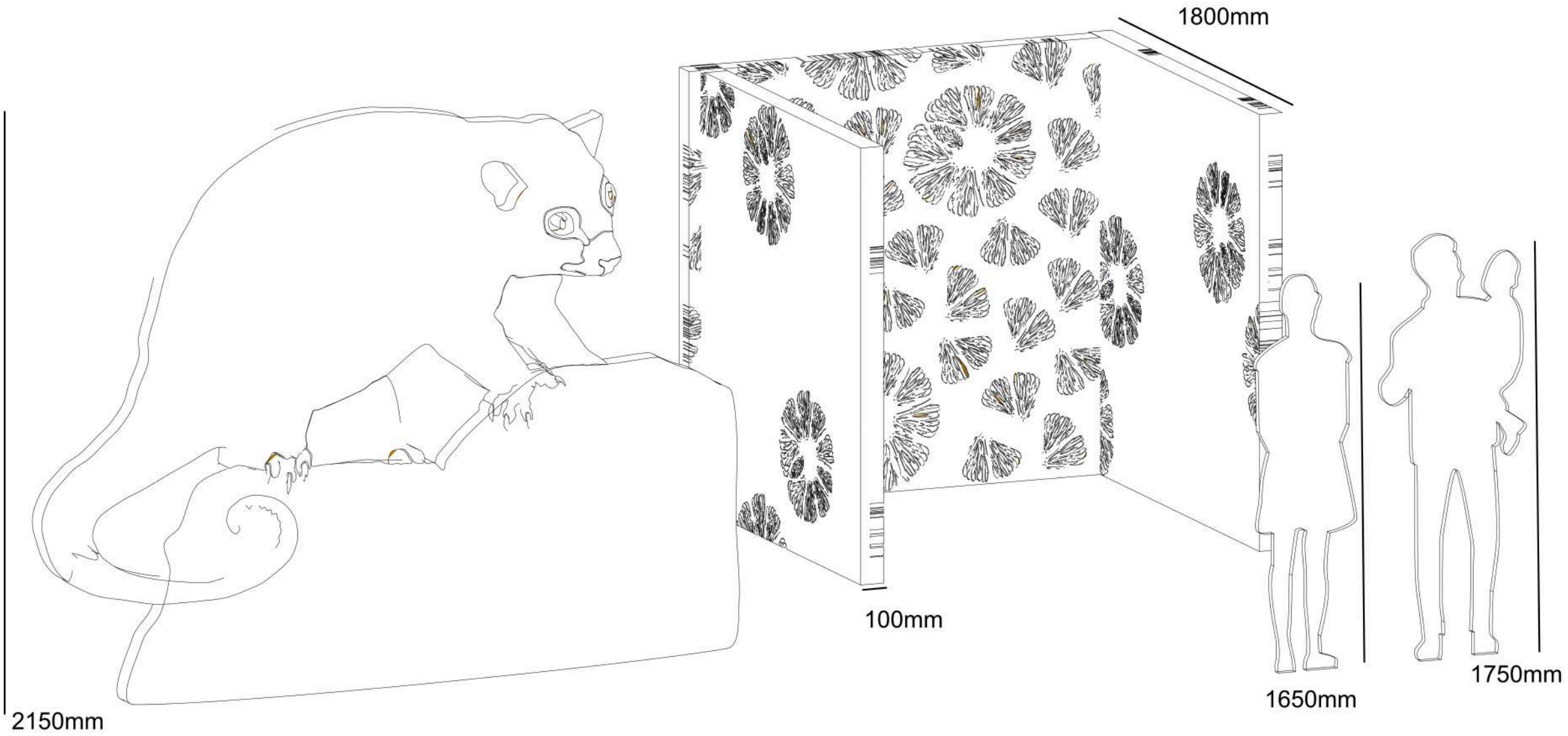
Exploded view mirrors the style of tech drawings exploded view. Whilst demonstraiting the detail of the oranges inner skin.

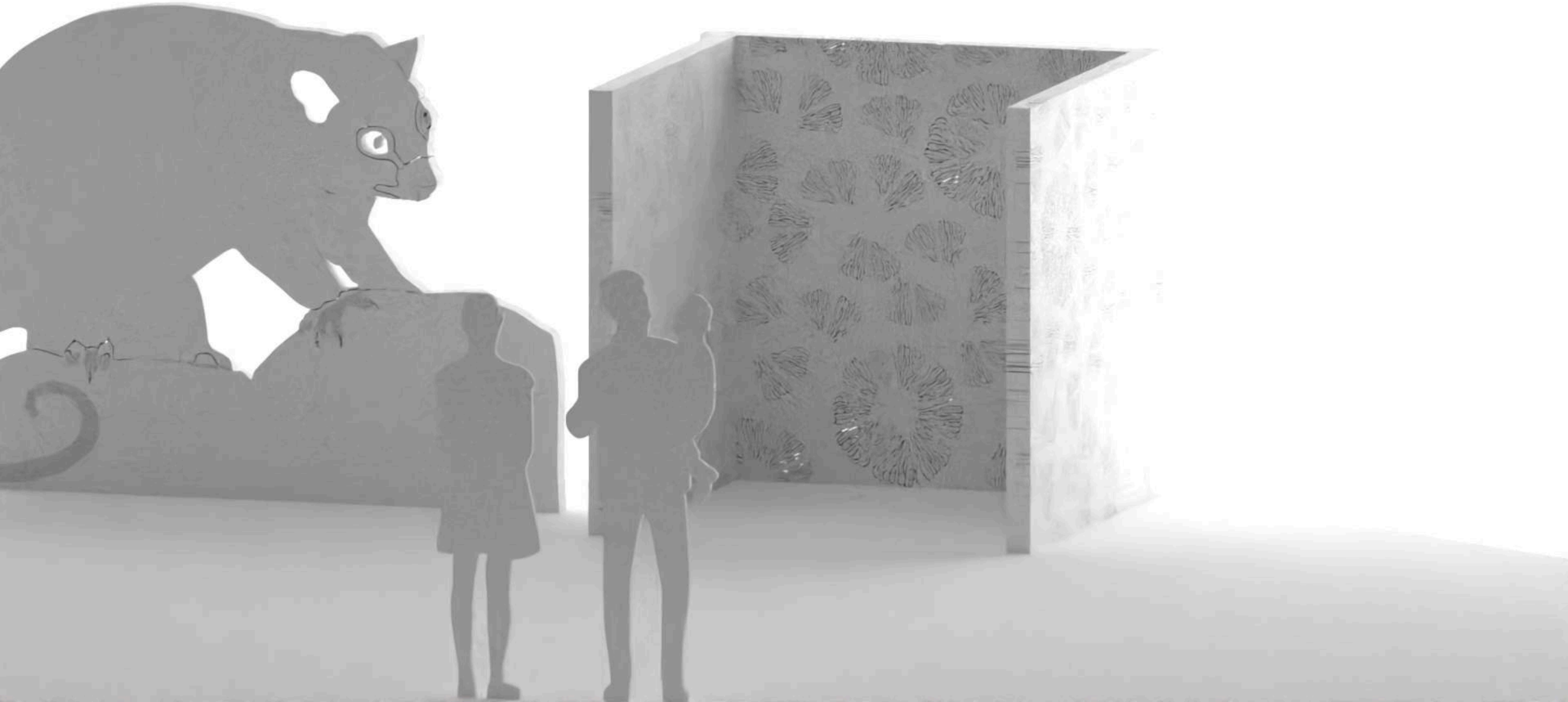


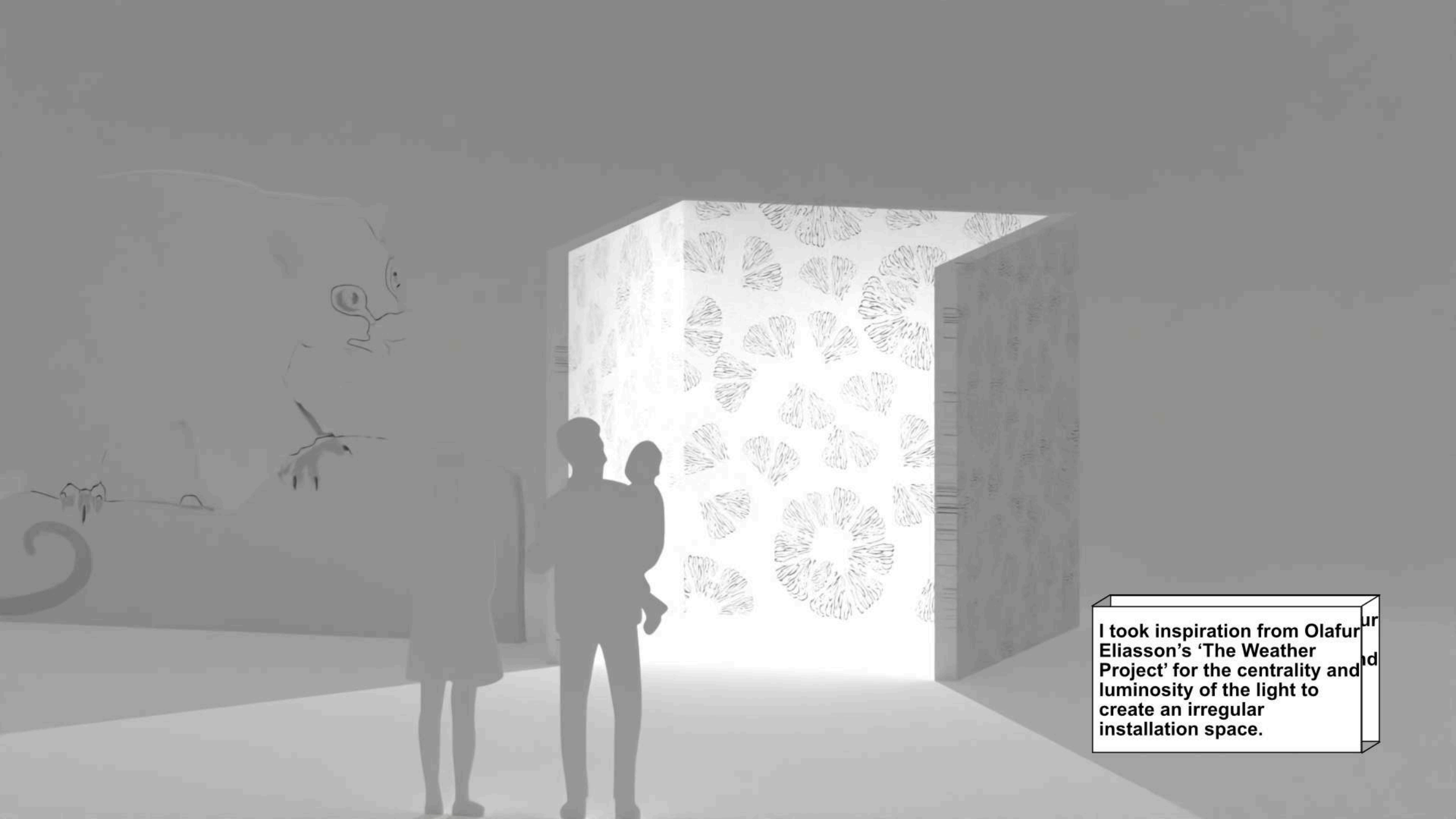
Tech drawing stylistic approach enhanced by adding the boarder and fill in boxes to mimic conventional engineering parts drawings.

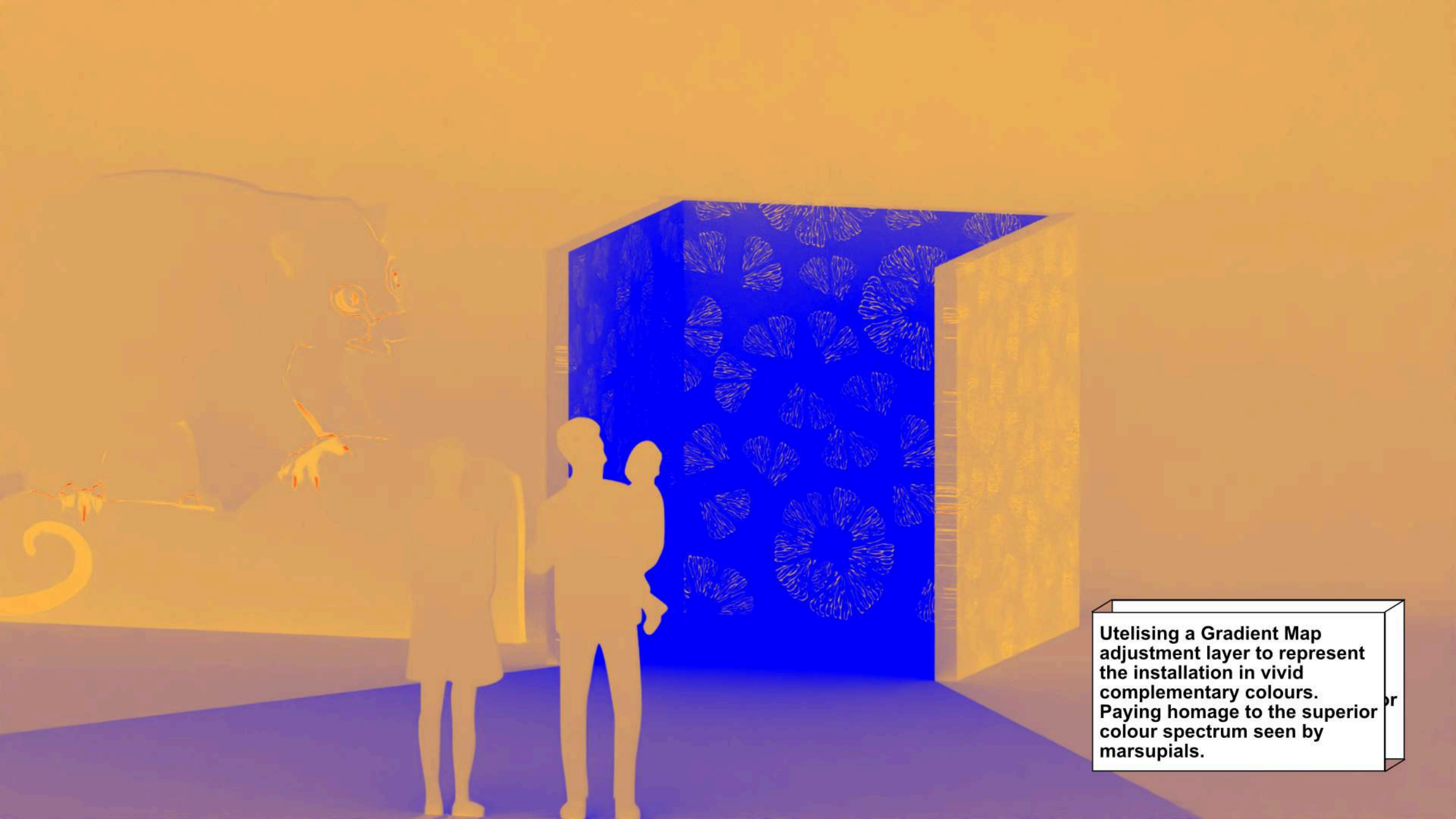


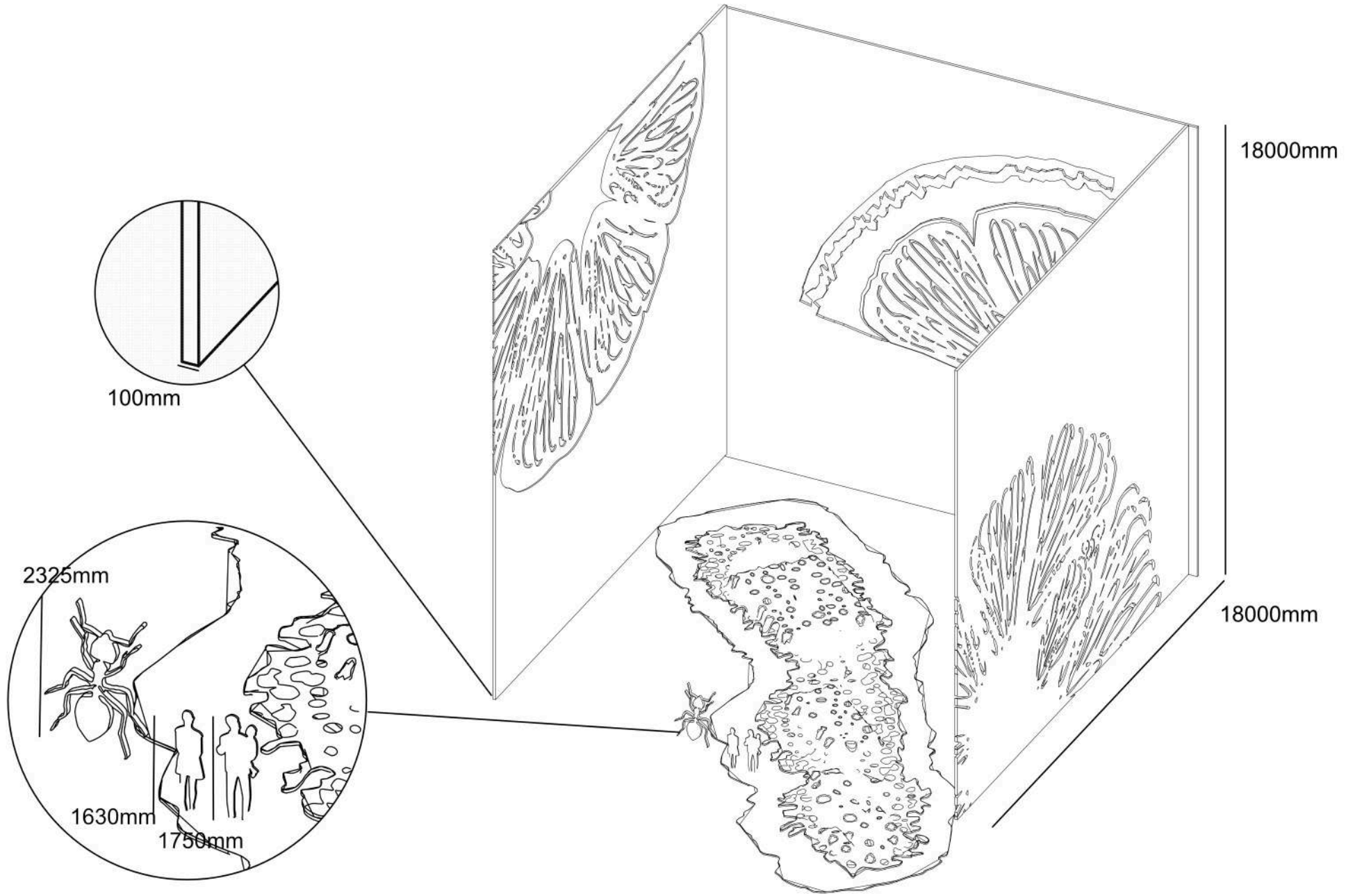



1800mm
100mm
2150mm
1650mm
1750mm

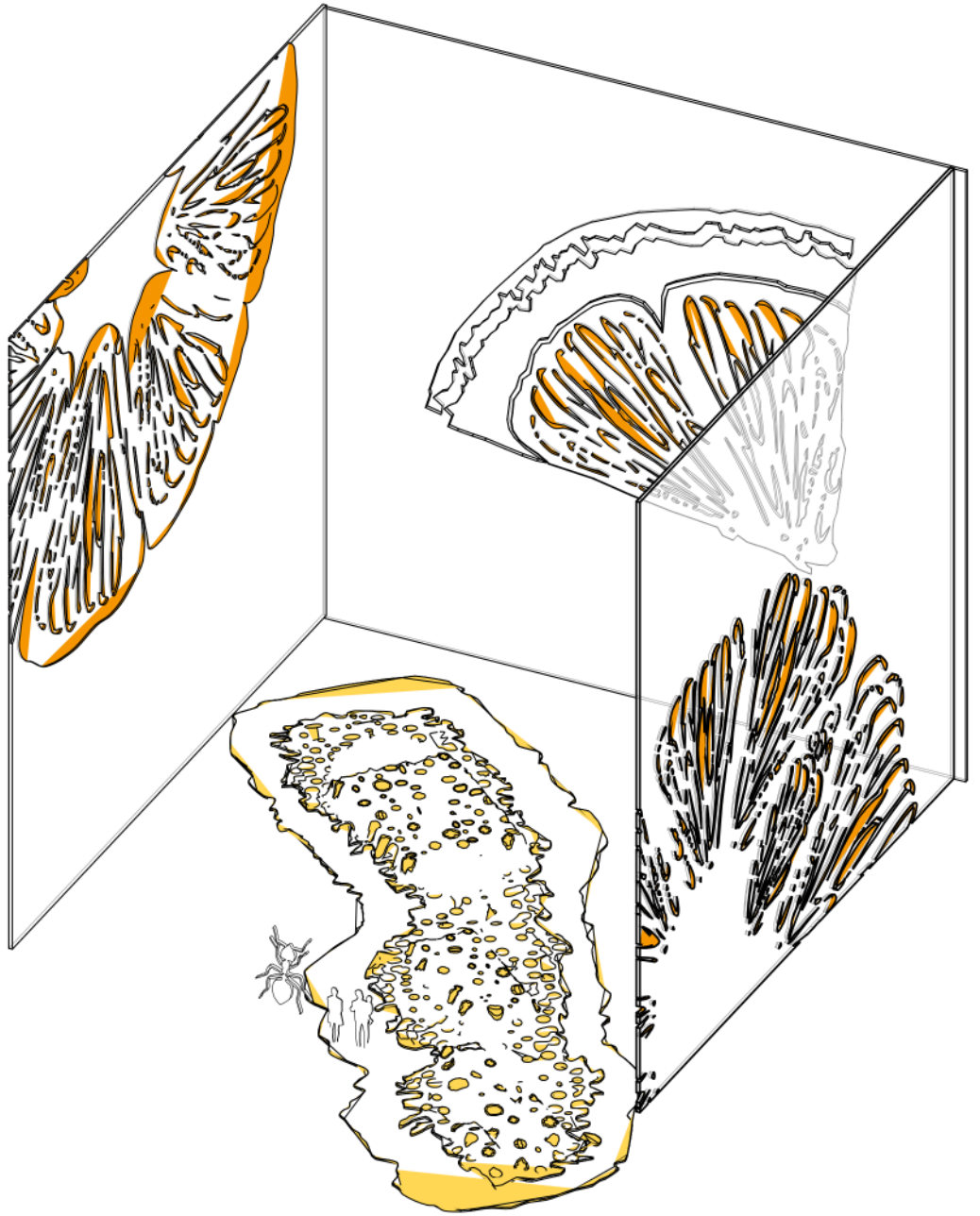



I took inspiration from Olafur Eliasson's 'The Weather Project' for the centrality and luminosity of the light to create an irregular installation space.

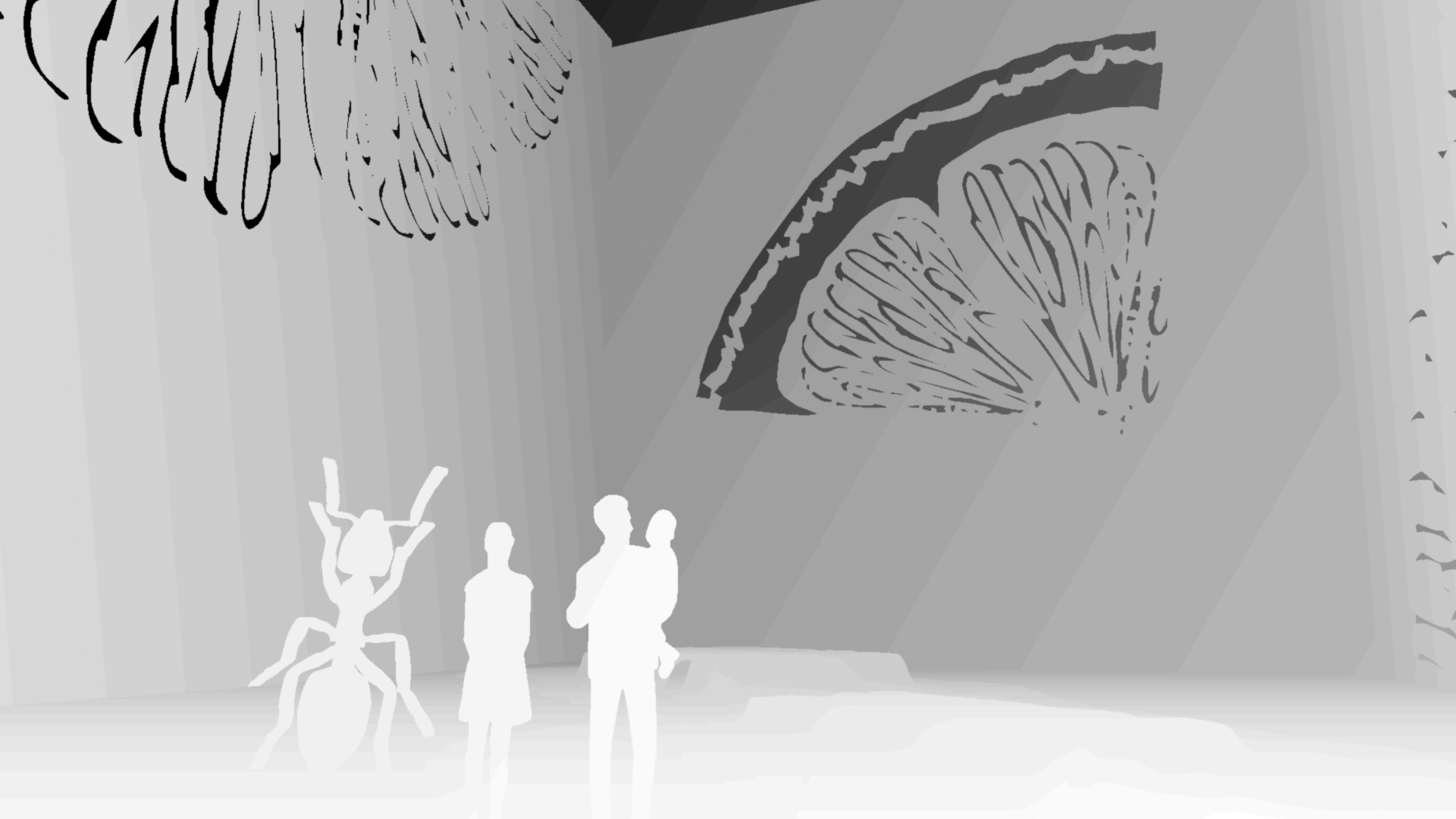

Utelising a Gradient Map adjustment layer to represent the installation in vivid complementary colours. Paying homage to the superior colour spectrum seen by marsupials.

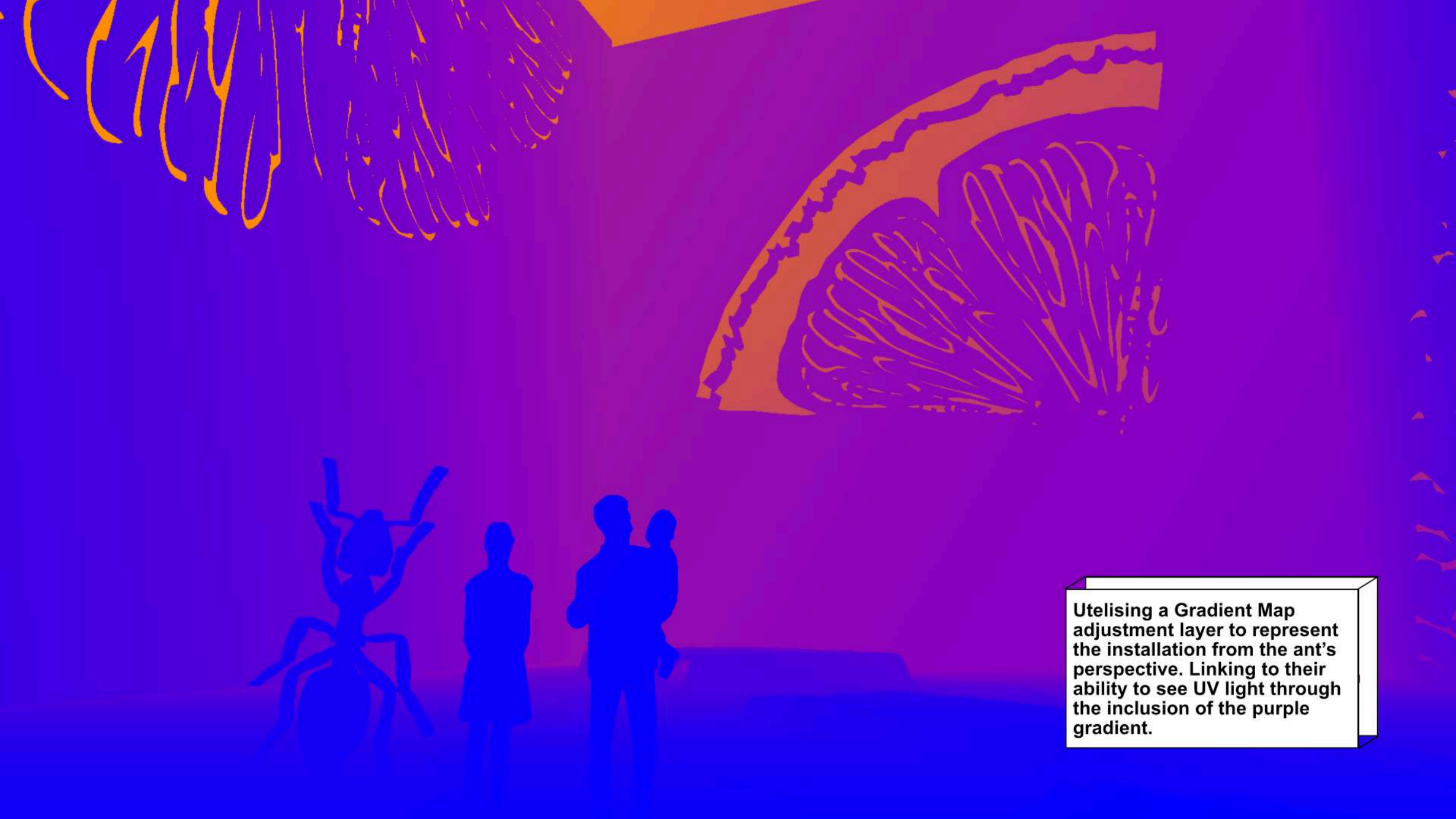

100mm
2325mm
1630mm
1750mm
18000mm
18000mm

I pushed this installation through the use of a mesh to represent the orange peel. The raised surface with the perforated holes creates an interactive exhibit for visitors to walk over and climb into the orange peel.

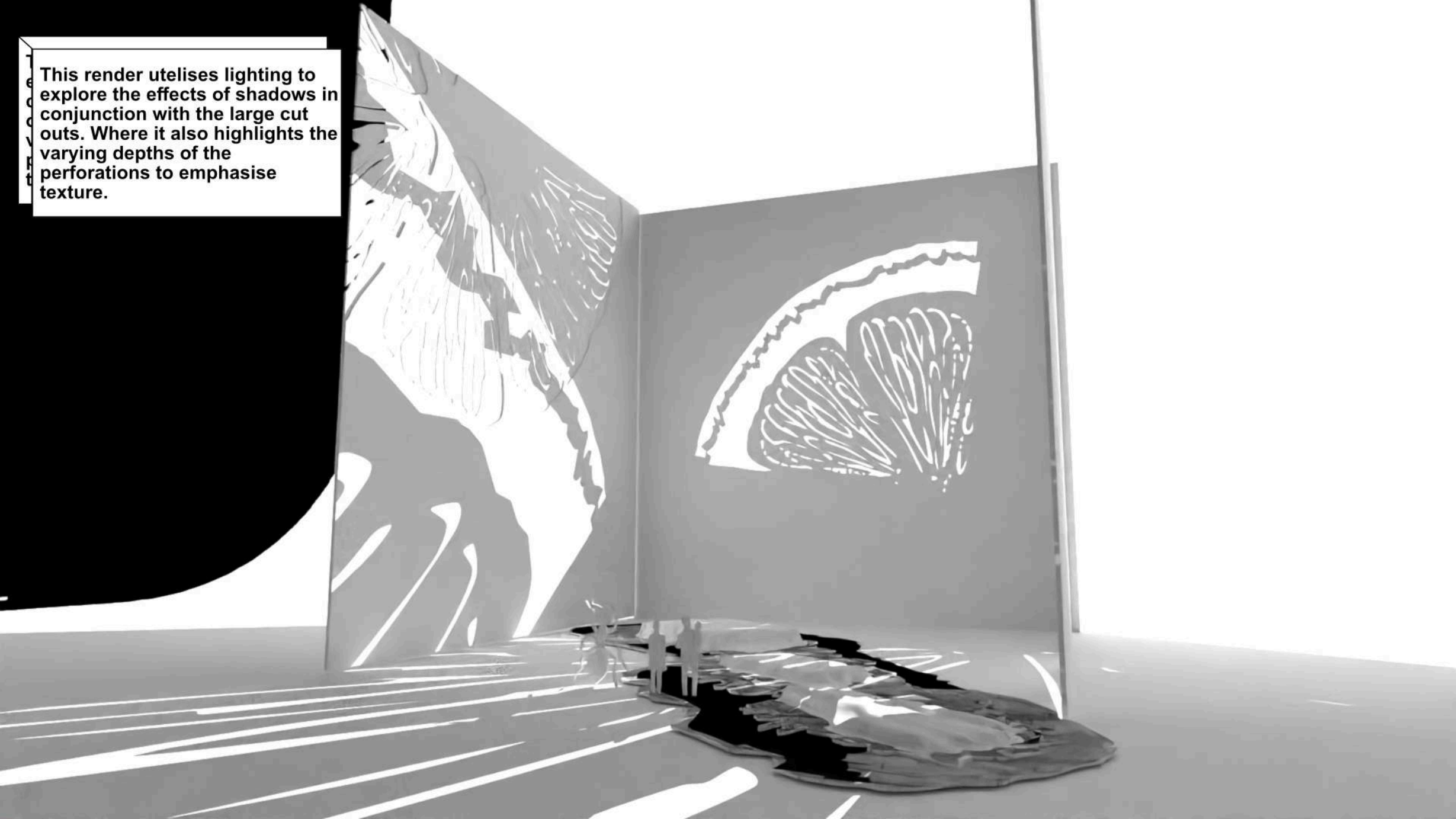







Utelising a Gradient Map adjustment layer to represent the installation from the ant's perspective. Linking to their ability to see UV light through the inclusion of the purple gradient.

This render utelises lighting to explore the effects of shadows in conjunction with the large cut outs. Where it also highlights the varying depths of the perforations to emphasise texture.

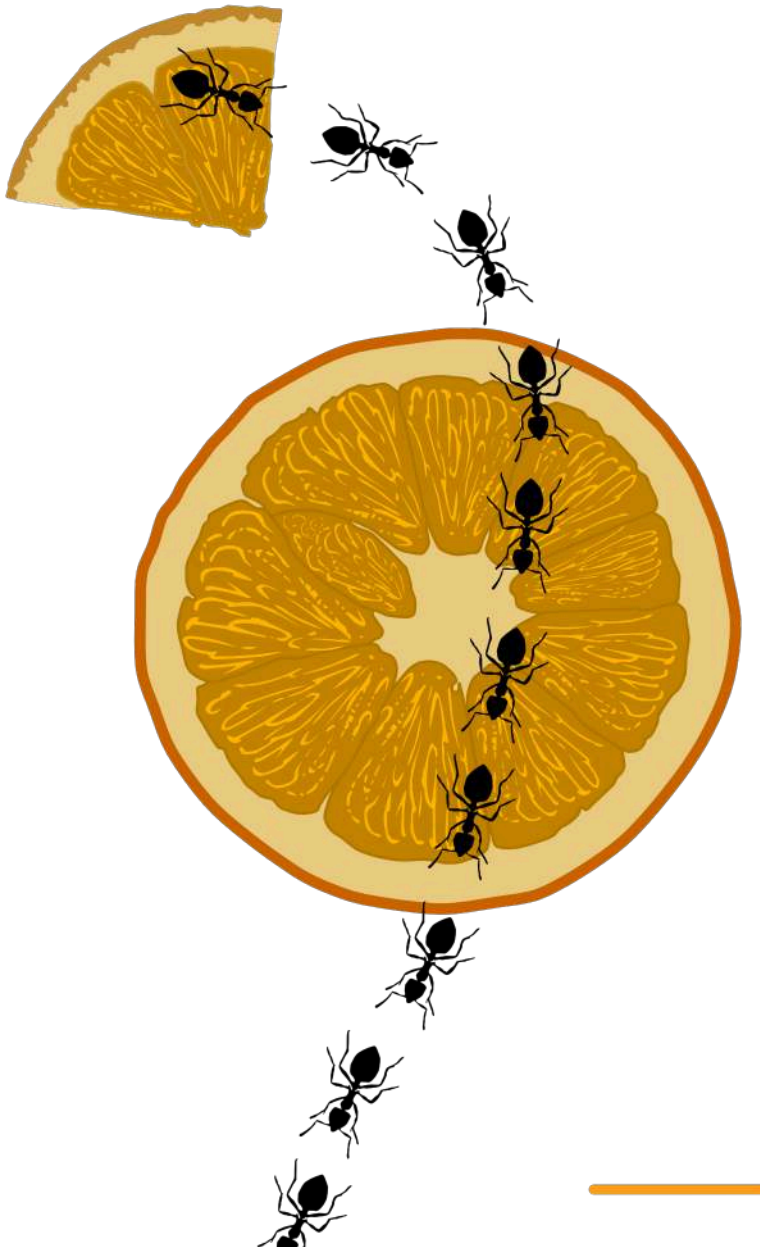


Imagine

Refection:

This assignment led to many peaks and troughs in terms of the learning curve, new concepts, varying tools and a deeper understanding of representation. Throughout this experience I learnt a lot about myself as a designer and have gained a more concrete understanding of the parts of design representation that I enjoy and thrive at as well as other aspect that require more attention.

My strengths within a design framework are within research and digital modelling. I enjoy diving into specific reasearch to support a design idea, hence why this presentation includes a wide range of points of reasearch. While this was my first experience with Rhino my previous work on Fusion 360 did make the process more intuitive. To tailor this project to this particular strength and area of interest for myself, I explored more tools in Rhino such as the mesh for the peel of the orange. This was implemented in conjunction with keeping the linework more simplistic as an area of design for me that requires more attentsion is both digital and hand drawing. This limited the detail within my modelled installation, leading to this assignement informing my decision to work on hand sketching and digital sketch in order to not limit future designs.



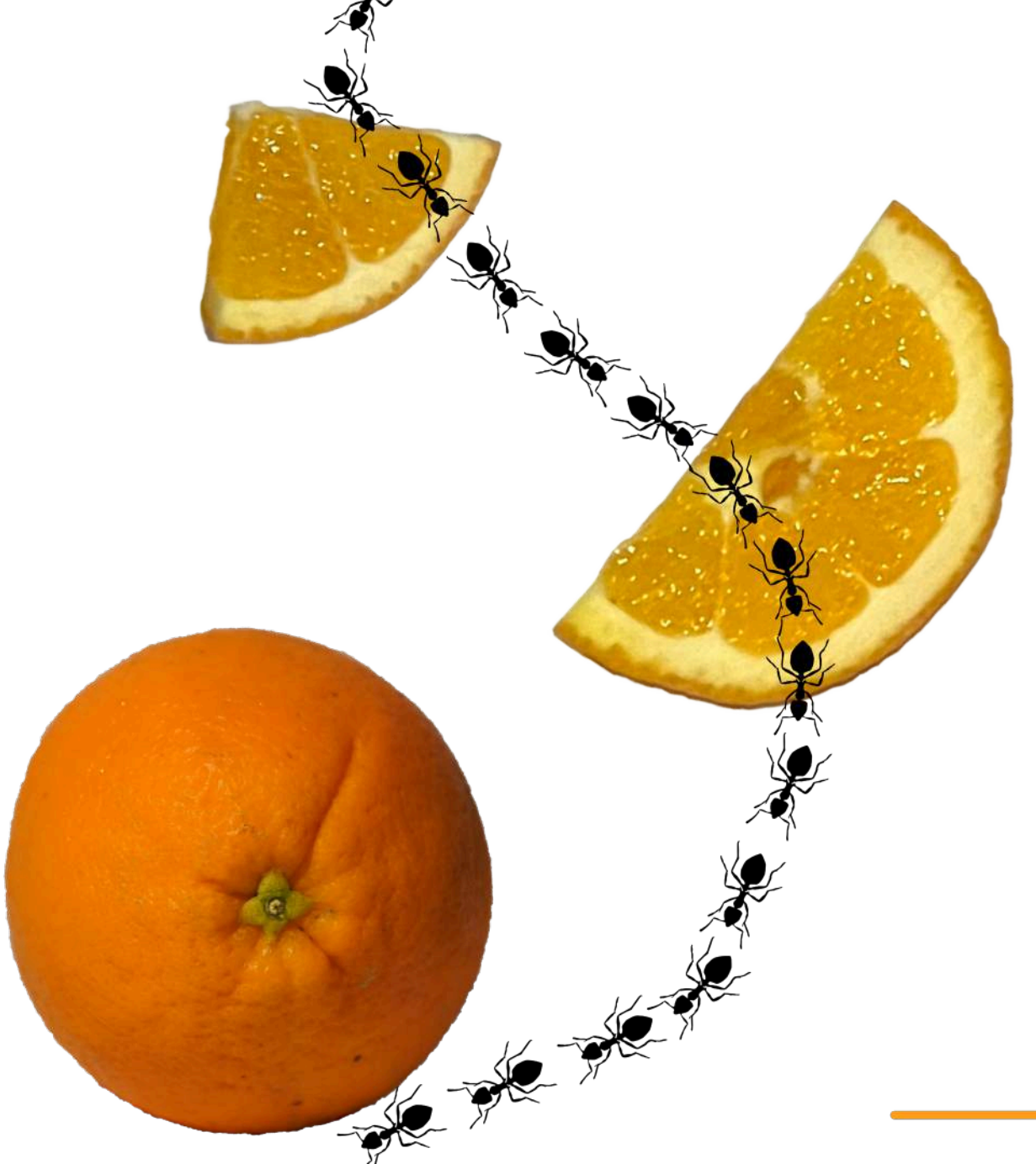
Imagine

Future:

Based on the content within my reflection reguarding this design and representation process I can surmise that my pathway within my discipline of engineering needs to align to my passion and strength within research and 3D modelling. This also informs some strategies that I will employ in future practise, including;

- Becoming well versed in the concept of a variety of software is more benefital then being optimised for one. Using affinity design, publisher and photo inconjunction with Rhino and DaVinci Resolve has set me up with a base understanding of different types of software that can be tranfered to other tools.

- Elevating visual representation of information through aesthetics as a valuable tool. The process of combining information stylisticly with infographics or visual aids lifts the quality of a presentation, as I learnt from formulating this presentation.



Resources

Slide 1:

- Podcast 1: The challenge: why should we study design representation?, Featuring :

  Host: Stanislav Roudavski

  Guest 1: Professor Hélène Frichot | Professor of Architecture and Philosophy, Director of the Bachelor of Design

  Guest 2: Dennis Prior, Senior Lecturer in Architectural Design, Pathway Coordinator for the discipline of architecture, Melbourne School of Design

  Guest 3: Christina Smith, Lecturer in Design, Victorian College of the Arts

- Podcast 2: The future: what will you work on in 20 years?

  Featuring:

  Host: Stanislav Roudavski

  Guest 1: Professor Dan Hill, Director, Melbourne School of Design

  Guest 2: Dr Dennis Prior, Senior Lecturer in Architectural Design, Pathway Coordinator for the discipline of architecture, Melbourne School of Design

  Guest 3: Christina Smith, Lecturer in Design, Victorian College of the Art

Slide 2:

- Video 1: Interview with Dale Ferguson, theatre designer. (Available on LMS)
- Image (Steampunk Pavilion)

  https://www.archdaily.com/926191/steampunk-pavilion-gwyllm-jahn-and-cameron-newnham-plus-soomeen-hahm-design-plus-igor-pantic

Slide 3:

- Video 2: Interview with Gwyllim Jahn, an architechural designer. (Available on LMS)
- Image (Death of a Salesman)

  https://cityhub.com.au/death-of-a-salesman-review/

Slide 4:

- Video 1: Interview with Dale Ferguson, theatre designer. (Available on LMS)
- Video 2: Interview with Gwyllim Jan, an architechural designer. (Available on LMS)
- Images

  Video 1: Interview with Dale Ferguson, theatre designer. (Available on LMS)

  Video 2: Interview with Gwyllim Jan, an architechural designer. (Available on LMS)

Slide 5:

- Images

  Coolpaste: https://packagingoftheworld.com/2019/03/coolpaste.html

  Christian Siriano's 2020 NYC Sketches: https://www.aol.com/lifestyle/nyfw-designers-share-inspiration-behind-151900692.html

Slide 6:

- Images

  Alessi Coffee Machine: https://www.elledecor.com/it/best-of/a27008852/coffee-maker-alessi-design-week-2019/

  Schiaparelli Sketch: https://www.vogue.com/fashion-shows/fall-2020-couture/schiaparelli/slideshow/collection#1

  Sewing Pattern: https://so-sew-easy.com/print-assemble-pdf-pattern/

  Venini Vase Concept Sketch: https://www.designboom.com/design/design-museum-holon-alessi-03-17-2015/

Slide 7:

- Images:

  Hermes Tech Drawing: https://www.charlottehofstetter.

Slide 8:

- Scientific information:

  https://en.wikipedia.org/wiki/Rutaceae

  https://en.wikipedia.org/wiki/Hesperidium

  https://www.researchgate.net/figure/Animal-species-consumers-of-citrus-fruit-part-of-the-fruit-consumed-and-geographical_tbl1_287815533

Slide 9:

- Cultural information:

  https://thenewgastronome.com/oranges/

  https://direct.mit.edu/jinh/article-abstract/50/1/129/49598/Golden-Fruit-A-Cultural-History-of-Oranges-in?redirectedFrom=fulltext

  https://www.researchgate.net/publication/325112964_Golden_Fruit_A_Cultural_History_of_Oranges_in_Italy

- Image:
  https://www.theoi.com/Titan/Hesperides.html

Slide 10:

- Pest information:

  https://citrusaustralia.com.au/wp-content/uploads/lemon_growing_manual_11_Pests.pdf

- Orange information:

  https://en.wikipedia.org/wiki/Citrus_%C3%97_sinensis

Slide 11:

- Orange parts information:

  https://www.researchgate.net/figure/Parts-of-an-orange-fruit-35_fig1_362468749

Slide 15:

- Historical information:

  https://storymaps.arcgis.com/stories/1a88363072674762b95e1ab4e7431fd0